On the Order of Privacy and Smoothing in Functional Mean Estimation

Abstract

Differential privacy (DP) provides a rigorous framework for protecting individual-level information while preserving data utility by adding a calibrated amount of noise to a summary statistic. Extending DP to functional data analysis (FDA) is challenging, as observations and target summaries take values in a potentially infinite-dimensional function space. While recent work has developed some privacy mechanisms for releasing functional summaries, a fundamental pipeline design question remains open: whether one should estimate the functional summary first and privatize it, or privatize the observed data first and subsequently fit the functional estimator. In this paper, we systematically investigate this ordering question for functional mean estimation by formulating and comparing two general pipelines—*Fit-Then-Protect* (FTP) and *Protect-Then-Fit* (PTF), respectively—under a common μ-Gaussian Differential Privacy (μ-GDP) budget. First, we establish a general μ-GDP mechanism for function-valued summaries in separable Banach spaces via compatible Gaussian process perturbations. Building upon this foundation, we derive finite-sample integrated mean squared error (IMSE) and empirical mean squared error (MSE) bounds for FTP and PTF in finite-dimensional orthonormal polynomial spaces and infinite-dimensional reproducing kernel Hilbert spaces (RKHSs). In the noiseless regime, we prove that exploiting functional structure prior to privatization allows FTP to strictly outperform PTF, achieving an asymptotic error ratio bounded by the reciprocal of the effective basis dimension of the estimated functional summary (namely, $1/m$ for m-dimensional polynomials and $1/T$ across T design points in RKHSs). Simulation studies validate these theoretical bounds and characterize how the constraint radius and observation noise govern the finite-sample trade-off between the two orderings.

Keywords: Gaussian differential privacy, functional data analysis, reproducing kernel Hilbert space, constrained optimization

1 Introduction

Differential privacy (DP) provides a rigorous framework for protecting individual-level information while balancing data usability. By carefully calibrating the noise added to the released summaries, DP blurs the contribution of an individual while preserving the utility of the data. Since Dwork et al. (2006) introduced DP, it has become a central paradigm in privacy-preserving data analysis, stimulating extensive research in statistics, machine learning, and related fields. Despite this progress, most existing DP methodology has been developed for finite-dimensional data and vector-valued statistics, leaving important challenges for more complex data objects.

These challenges are particularly relevant in Functional Data Analysis (FDA), where observations are inherently infinite-dimensional. FDA is a statistical framework for analyzing data whose basic observational unit is a function or curve rather than a finite-dimensional vector. Typical examples include growth curves, wearable sensor data, and other densely or continuously recorded signals (Wang et al., 2016; Ramsay and Silverman, 2005; Acar-Denizli

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and Delicado, 2025). Instead of treating observations at different time points independently, FDA models each subject's data as a smooth function, allowing statistical procedures to exploit the within-curve correlation and geometric structure inherent in functional observations. Over the past three decades, FDA has become an important area of modern statistics with applications across biology, medicine, engineering, and environmental science.

As functional data become increasingly prevalent, protecting the privacy of function-valued information has become an important challenge. In many applications, the object of interest is itself a function—for example, a mean trajectory, a regression function, or an estimated individual curve—which may reveal sensitive information about individuals. While classical DP mechanisms are primarily designed for finite-dimensional statistics, extending privacy guarantees to infinite-dimensional function spaces is substantially more challenging, requiring both suitable notions of sensitivity and carefully constructed noise processes. These challenges naturally raise the question of how DP should be formulated and implemented for function-valued outputs. Previous work has built (ϵ, δ)-DP for functionals in different function spaces (Hall et al., 2013; Mirshani et al., 2019), while recent work extends this to the Gaussian Differential Privacy (GDP) framework under Reproducing kernel Hilbert space (RKHS) (Räisä et al., 2024).

Another fundamental question arises regarding where privacy should be enforced, even when privacy mechanisms for functional outputs are available. Given a functional estimation problem, privacy can be incorporated in at least two natural ways. One may first estimate the mean function and then privatize the resulting estimate, or alternatively privatize the data first and subsequently perform the estimation using the protected observations. We refer to these two strategies as *Fit-Then-Protect* (FTP) and *Protect-Then-Fit* (PTF), respectively. Although both procedures can be calibrated under a common privacy budget after appropriate calibration, they exploit the functional structure of the data in fundamentally different ways, and little is known about when one ordering is statistically preferable to the other.

In this paper, we investigate the above question under the framework of GDP for functional mean estimation. We aim to understand how different privacy-estimation orderings influence statistical efficiency under a common privacy budget. We first establish a GDP mechanism for releasing function-valued summaries in general Banach spaces, providing a sufficient condition under which Gaussian process perturbations achieve a prescribed GDP guarantee. Building upon this result, we study the statistical properties of the FTP and PTF pipelines under a common privacy budget. We analyze the sensitivity of the corresponding mechanisms together with the mean squared error of the resulting private estimators in several function spaces, including polynomial space and RKHS, and characterize the scenarios in which each strategy is preferable.

Contributions: Our main contributions are summarized as follows:

- We formulate the problem of differentially private functional mean estimation from the perspective of privacy-estimation ordering. In particular, we identify two general pipelines, Fit-Then-Protect (FTP) and Protect-Then-Fit (PTF), and provide a unified framework for comparing them under a common privacy budget. To the best of our knowledge, this is the first work to systematically study how the ordering between functional estimation and privacy perturbation affects statistical accuracy.

- We establish a GDP mechanism for function-valued summaries in Banach spaces. Building upon Gaussian perturbation mechanisms previously developed for (ϵ, δ)-DP, we derive sufficient conditions under which Gaussian process perturbations satisfy a prescribed GDP guarantee. This provides the theoretical foundation for applying GDP directly to infinite-dimensional function-valued summaries.

- Under a common privacy budget, we derive theoretical comparisons between the FTP and PTF pipelines without observation noise. In particular, we establish upper bounds for their integrated mean squared error (IMSE) and mean squared error (MSE) in polynomial spaces with $L^2[0,1]$ norm and in RKHS with Gaussian kernels, where FTP provably achieves strictly smaller estimation error than PTF. Our analysis quantifies how exploiting functional structure prior to privatization can substantially improve statistical efficiency.

Related Work: Previous work on differential privacy for function-valued objects has primarily focused on extending classical privacy mechanisms from finite-dimensional vectors to infinite-dimensional function spaces. Hall et al. (2013) proposed one of the earliest Gaussian process perturbation mechanisms for releasing functional summaries while satisfying (ϵ, δ)-DP in RKHS. Building upon this framework, Mirshani et al. (2019) established a general theory of Gaussian perturbation mechanisms in Banach spaces and characterized sufficient conditions under which function-valued summaries satisfy (ϵ, δ)-DP. Complementary work studied the limitations of elliptical perturbations in infinite-dimensional settings (Reimherr and Awan, 2019) and extended the exponential mechanism to functional summaries, including functional principal component analysis (Awan et al., 2019). More recently, Räisä et al. (2024) established GDP guarantees for a Gaussian-process-based functional mechanism in RKHS. Beyond general privacy mechanisms, Xue et al. (2026) studied private distributed FDA and established minimax results for functional mean and varying-coefficient estimation. These studies primarily develop privacy mechanisms or estimators under prescribed privacy constraints, leaving open how the ordering of privacy perturbation and functional estimation affects statistical accuracy.

Another line of research investigates the role of dependence and structural information in privacy mechanisms. Cao et al. (2019) analyzed privacy leakage under temporal correlations in continuous data release, while other work has developed correlated-noise mechanisms to preserve time-series structure and and mitigate vulnerabilities of independent perturbations (Wang and Xu, 2017; McElroy et al., 2023). More broadly, geometry-aware and structured mechanisms have been shown to achieve near-optimal or improved utility over conventional independent perturbations in finite-dimensional settings, including mechanisms for vector- and matrix-valued queries, private learning (Nikolov et al., 2013; Awan and Slavković, 2021; Chanyaswad et al., 2018; Choquette-Choo et al., 2024).

Organization: The remainder of this paper is organized as follows. Section 2 reviews the necessary background on differential privacy and functional data analysis. Section 3 formulates the problem of differentially private functional mean estimation and introduces the *Fit-Then-Protect* (FTP) and *Protect-Then-Fit* (PTF) strategies. Section 4 establishes a Gaussian Differential Privacy mechanism for function-valued summaries and develops theoretical comparisons between the two strategies. Section 5 presents simulation studies

evaluating their empirical performance under different function spaces. Section 6 concludes the paper with a discussion of potential extensions.

2 Background

In this section, we review the fundamental and necessary concepts and results of DP and FDA related to our FTP and PTF mechanisms. In Section 2.1, we first discuss the trade-off function and the definition of μ-GDP, then introduce the definition of sensitivity and the Gaussian mechanisms to ensure μ-GDP, and show existing μ-GDP properties on composition and post-processing. In Section 2.2, we give the definition and notations of some common function spaces in FDA, such as Banach space, Reproducing kernel Hilbert space (RKHS), and then define the Gaussian process in a general Banach space, and introduce the corresponding Cameron-Martin space and discuss its relationship with RKHS.

2.1 Differential Privacy

Let a dataset $X = (x_1, \ldots, x_n)$ denote a collection of n records, each $x_i \in \mathcal{X}$ for some set $\mathcal{X}$, and $\mathcal{X}^n$ denotes the collection of all possible data sets. For two data sets $X, X' \in \mathcal{X}^n$, we denote $X \sim X'$ if they differ in at most one entry, that is, their *Hamming distance* $d_H(X, X') = \sum_{i=1}^n \mathbf{1}\{x_i \neq x_i'\} \leq 1$. We call such X and X' as *adjacent (or neighboring) data sets*. A *mechanism* $\mathcal{M} : \mathcal{X}^n \to \mathcal{Y}$ is a randomized algorithm taking a data set X as input and taking values on a measurable space $\mathcal{Y}$. At a high level, differential privacy quantifies the difference between outputs of the mechanism when taking two adjacent data sets as inputs.

We use the μ-GDP framework throughout the paper, where the difference is measured through hypothesis testing. The intuition is that the more difficult it is to distinguish between the mechanism's outputs on two adjacent datasets, the stronger the privacy guarantee it provides. Let P and Q denote the probability distributions of $\mathcal{M}(X)$ and $\mathcal{M}(X')$, respectively. Consider a hypothesis test between: $H_0 : Y \sim P$, $H_1 : Y \sim Q$, where Y is the observed output. Let $0 \leq \psi \leq 1$ be a rejection rule given Y from either P or Q. We use $\alpha_\psi = \mathbb{E}_P[\psi]$ to denote the Type I error, and $\beta_\psi = 1 - \mathbb{E}_Q[\psi]$ to denote the Type II error, and we measure the difficulty of this test via the *trade-off function* defined below.

Definition 1 (Trade-off function (Dong et al., 2022)) *For any two probability distributions P and Q on the same space. Define the trade-off function $T(P, Q) : [0, 1] \to [0, 1]$ as*

$$T(P, Q)(\alpha) = \inf_{\psi} \{\beta_\psi : \alpha_\psi \leq \alpha\},$$

where ψ ranges over all measurable rejection rules.

The trade-off function characterizes the minimal type II error as a function of the prescribed Type I error. A larger trade-off function indicates that the two distributions are more difficult to distinguish, corresponding to a stronger privacy guarantee when P and Q arise from neighboring data sets. For $\mu \geq 0$, define $G_\mu = T\big(N(0, 1), N(\mu, 1)\big)$ to be the *Gaussian trade-off function* between unit-variance Gaussian with mean difference μ. We can therefore define Gaussian differential privacy.

Definition 2 (μ-Gaussian Differential Privacy (GDP), Dong et al., 2022) *A mechanism $\mathcal{M}$ satisfies μ-GDP if for all adjacent data sets $X \sim X' \in \mathcal{X}^n$,*

$$T(\mathcal{M}(X), \mathcal{M}(X')) \geq G_\mu.$$

Intuitively, a mechanism $\mathcal{M}$ satisfies μ-GDP means that testing $H_0 : Y \sim \mathcal{M}(X)$ versus $H_1 : Y \sim \mathcal{M}(X')$ is at least as hard as testing $H_0 : Y \sim N(0,1)$ versus $Y \sim N(\mu, 1)$. The parameter μ quantifies the distinguishability between the outputs of a mechanism on adjacent data sets: smaller values of μ correspond to stronger privacy guarantees, but with less data usability. An important advantage of GDP is that it provides an interpretable, one-parameter characterization of privacy and admits convenient composition rules.

Definition 3 (Global sensitivity in a normed space) *Let $(\mathcal{B}, \|\cdot\|_\mathcal{B})$ be a normed vector space, and let $S : \mathcal{X}^n \to \mathcal{B}$ be a statistic computed from a dataset $X \in \mathcal{X}^n$. The global sensitivity of S with respect to $\|\cdot\|_\mathcal{B}$ is defined as*

$$\Delta_\mathcal{B}(S) = \sup_{X \sim X'} \|S(X) - S(X')\|_\mathcal{B},$$

The sensitivity measures the maximum change in the statistic in the norm of the output space when one individual's data is modified. In finite-dimensional settings when $\mathcal{B} = \mathbb{R}^d$, ℓ_1- and ℓ_2-sensitivities, denoted as Δ_1 and Δ_2, are two common choices. Definition 3 extends naturally to function-valued statistics taking values in a function space, while it may require some specific norm to satisfy the desired DP property.

In practice, sensitivity may be unbounded when the data are unbounded, making it difficult to calibrate noise for privacy mechanisms. *Clipping* is a standard technique in DP for limiting the influence of individual data values. The clipping of $x \in \mathbb{R}$ to the interval $x \in \mathbb{R}$ is defined as

$$\text{clip}_{[a,b]}(x) = \min\{b, \max\{a, x\}\}.$$

By restricting values to a bounded interval, clipping can help control the sensitivity of statistical queries and facilitate the calibration of noise required to achieve certain DP guarantee.

Proposition 4 (The Gaussian Mechanism (Dong et al., 2022)) *Let $S : \mathcal{X}^n \to \mathbb{R}^d$ be a function with ℓ_2 sensitivity, $\Delta_2(S) = \sup_{X \sim X'} \|S(X) - S(X')\|_2$. For any S, the Gaussian Mechanism releases*

$$\mathcal{M}(X) = S(X) + Z,$$

where $Z \sim N(0, \sigma^2 I_d)$. Then $\mathcal{M}(X)$ satisfies μ-GDP if $\sigma = \Delta_2(S)/\mu$.

The Gaussian mechanism provides a fundamental approach to achieving GDP by adding Gaussian noise calibrated to the ℓ_2-sensitivity of the underlying statistic in finite-dimensional settings. For function-valued statistics, Gaussian noise vector is replaced by a Gaussian random element taking values in an appropriate function space, often represented by a Gaussian process. In this setting, the privacy guarantee depends not only on the sensitivity of the statistic but also on the properties of the Gaussian noise process and its associated function space. We introduce the necessary functional-analytic framework and develop the corresponding function-valued mechanisms in Section 2.2 and Section 4.

Proposition 5 (Composition of GDP (Dong et al., 2022)) *Suppose independent mechanisms $\mathcal{M}_1, \ldots, \mathcal{M}_k$ operate on the same data set and satisfying μ_1-GDP, $\ldots$, μ_k-GDP, respectively. Then the composed mechanism $\mathcal{M} = (\mathcal{M}_1, \ldots, \mathcal{M}_k)$ satisfies μ-GDP with $\mu = \sqrt{\mu_1^2 + \cdots + \mu_k^2}$.*

Proposition 6 (Post-Processing) *Suppose a mechanism $\mathcal{M} : \mathcal{X}^n \to \mathcal{Y}$ satisfies μ-GDP, and let $g : \mathcal{Y} \to \mathcal{R}$ be any measurable function. Define $\mathcal{M}'(X) = g(\mathcal{M}(X))$. Then $\mathcal{M}'$ satisfies the same privacy guarantee.*

The composition property characterizes how privacy guarantees accumulate when multiple mechanisms are applied to the same dataset, while the post-processing property ensures that data-independent transformations of privatized outputs incur no additional privacy loss. These properties are useful for analyzing the PTF mechanisms, particularly when combining multiple private releases or applying subsequent estimation procedures to privatized data.

2.2 Functional Data Analysis

To discuss functions, we first introduce a general space. Let $\mathbb{B}$ be a Banach space, which is a complete normed vector space. We can define linear functional $f : \mathbb{B} \to \mathbb{R}$ with $\|f\|_{\mathbb{B}^*} < \infty$, where $\|f\|_{\mathbb{B}^*} = \sup_{\|h\|_{\mathbb{B}} \leq 1} |f(h)|$. The collection of these functionals $\mathbb{B}^* = \{f : \mathbb{B} \to \mathbb{R} \mid f \text{ is linear and } \|f\|_{\mathbb{B}^*} < \infty\}$ forms the dual space of $\mathbb{B}$. Note that $\mathbb{B}^*$ under sup norm is complete, that is, $\mathbb{B}^*$ is also a Banach space (Fabian et al., 2011).

We say that a random vector is normally distributed with mean μ and covariance C, and denote it as $N(\mu, C)$, the Gaussian distribution over $\mathbb{R}$. We can extend this notation for a general underlying space and include subscripts for other spaces, for example, $\mathcal{N}_{\mathbb{B}}$ over $\mathbb{B}$.

Definition 7 (Gaussian process) *A random process $Z \in \mathbb{B}$ is called Gaussian if $f(Z)$ is Gaussian in real space, for any continuous linear functional $f \in \mathbb{B}^*$.*

A Gaussian process Z is uniquely parametrized by a mean $\mu \in \mathbb{B}$ and a covariance operator

$$\mathrm{Cov}(f(Z), g(Z)) = \int f(Z - \mu) g(Z - \mu) d\nu(Z),$$

where $f, g : \mathbb{B} \to \mathbb{R}$ are functionals and $\nu(Z)$ is the probability measure over $\mathbb{B}$ induced by Z. When $f \in \mathbb{B}^*$ we have the following property:

$$f(Z) \sim N(f(\mu), \mathrm{Cov}(f(Z), f(Z))).$$

Definition 8 (Cameron–Martin Space) *Let $Z \sim \mathcal{N}_{\mathbb{B}}(0, C)$ be a centered Gaussian random element. The Cameron–Martin space $\mathcal{H}_C$ is the Hilbert space of directions $h \in \mathbb{B}$ for which translation by h preserves the equivalence class of the Gaussian measure induced by Z. Its inner product is denoted by $\langle \cdot, \cdot \rangle_{\mathcal{H}_C}$ and its norm by $\| \cdot \|_{\mathcal{H}_C}$.*

Definition 9 (Reproducing Kernel Hilbert Space) *Let $\mathcal{T}$ be an index set and let $K : \mathcal{T}\times\mathcal{T} \to \mathbb{R}$ be a symmetric positive semidefinite kernel. The reproducing kernel Hilbert space (RKHS) associated with K, denoted by $\mathcal{H}_K$, is the Hilbert space of functions $f : \mathcal{T} \to \mathbb{R}$ satisfying the reproducing property*

$$f(t) = \langle f, K(t,\cdot)\rangle_{\mathcal{H}_K}, \qquad f \in \mathcal{H}_K, \quad t \in \mathcal{T}.$$

In particular, by the Cauchy–Schwarz inequality, $|f(t)| \le \|f\|_{\mathcal{H}_K}\sqrt{K(t,t)}$. Thus, if $\|f\|_{\mathcal{H}_K} \le B$, then $|f(t)| \le B\sqrt{K(t,t)}$. This reproducing property provides a direct connection between the RKHS norm and the pointwise sensitivity of function-valued statistics.

Remark 10 (RKHS and Cameron–Martin Space) *The RKHS and Cameron–Martin space are distinct concepts in general, but they coincide in the standard Gaussian-process construction associated with a covariance kernel. In particular, when a mean-zero Gaussian process with kernel K is realized as a Gaussian random element in an appropriate function space (e.g. $\mathbb{B}$), the corresponding Cameron–Martin space is the RKHS $\mathcal{H}_K$.*

Definition 11 (Sensitivity for functionals.) *When the target quantity is a function, $f : \mathcal{X} \to \mathcal{H}$, where $\mathcal{H}$ is a function space (e.g. a Banach space or RKHS), sensitivity depends on the norm imposed on $\mathcal{H}$:*

$$\Delta_{\mathcal{H}}(f) = \sup_{X\sim X'} \|f(X) - f(X')\|_{\mathcal{H}}.$$

Example 1 *The following two are the common spaces in FDA.*

- *If $\mathcal{H} = L^2[0,1]$, then*
$$\|f\|_{L^2} = \left(\int_0^1 f(t)^2 dt\right)^{1/2}.$$
- *If $\mathcal{H}$ is an RKHS with kernel K, i.e. $\mathcal{H}_K$, then the RKHS norm $\|\cdot\|_{\mathcal{H}_K}$ is used.*

The choice of norm is important because it determines both the geometry of the function space and the amount of noise needed for privacy.

3 Problem Setup

In this section, we first specify our functional data model with necessary assumptions, then we define the privacy and estimation operators and formalize the proposed two possible orderings of fitting and privatizing: *Fit-Then-Protect* (FTP) and *Protect-Then-Fit* (PTF) with algorithmic implementation in the general Hilbert space $\mathcal{H}$. Our goal is to release a private estimate of the sample mean function under a prescribed GDP guarantee. Finally we discuss the two metrics, IMSE and MSE for evaluating the two procedures of FTP and PTF and the corresponding spaces for each metric.

3.1 Functional Data Model

Throughout the paper, the observed data consists of n independent functional trajectories $\mathbf{x}_i = (x_{i1}, \ldots, x_{iT})^\top \in \mathcal{X}$ at common and distinct points $\mathbf{t} = (t_1, \ldots, t_T)^\top \in \mathcal{T}$. We set $\mathcal{T} = [0,1]$, $\mathcal{X} = \mathbb{R}^T$. Denote $X = (\mathbf{x}_1, \ldots, \mathbf{x}_n)^\top \in \mathbb{R}^{n \times T}$ as the observed data matrix. Let $f_1, \ldots, f_n \in \mathcal{H}$ denote the latent functional trajectories in a general Hilbert space $\mathcal{H}$, and let $\mathbf{f}_i = (f_i(t_1), \ldots, f_i(t_T))^\top$ denote the evaluation vector of f_i at $\mathbf{t}$. Let $\epsilon_1, \ldots, \epsilon_n \in \mathbb{R}^T$ denote the random observation noise. The functional data are observed according to

$$\mathbf{x}_i = \mathbf{f}_i + \epsilon_i, \quad i = 1, \ldots, n \tag{1}$$

with the following assumptions:

A.1 (Bounded functional norm) $\|f_i\|_{\mathcal{H}} \leq B, i = 1, \ldots, n$ for some unknown constant $B > 0$.

A.2 (Gaussian observation noise) $\epsilon_1, \ldots, \epsilon_n \overset{i.i.d.}{\sim} N(0, \sigma_0^2 I_T)$. $\sigma_0 \geq 0$ is fixed and unknown.

Assumptions specific to polynomial spaces and RKHS will be stated in the corresponding subsections. Our objective is to estimate the sample mean function $\bar{f} = \frac{1}{n}\sum_{i=1}^n f_i$, while satisfying a prescribed μ-GDP guarantee.

3.2 Privacy–Estimation Operators and Orderings

To estimate an individual function from its observed trajectory, we define a function estimation operator $\mathcal{F} : \mathcal{X} \to \mathcal{H}$, given by

$$\mathcal{F}(\mathbf{x}) = \underset{f \in \mathcal{H}}{\operatorname{argmin}} \ \mathcal{L}(f; \mathbf{x}) \quad \text{subject to} \quad \|f\|_{\mathcal{H}} \leq B', \tag{2}$$

where $\mathcal{L}$ denotes a prescribed empirical loss and B' is the fitting radius we select. For our functional data model, $\mathcal{L}$ also depends on t_j's, we omit this as they are treated as given across the paper. For example, weighted square loss

$$\mathcal{L}(f; \mathbf{x}) = (\mathbf{f} - \mathbf{x})^\top W (\mathbf{f} - \mathbf{x}), \quad \mathbf{f} = (f(t_1), \ldots, f(t_T))^\top \tag{3}$$

is often considered, which depends on the choice of the weight matrix W. A common choice is to set the weight matrix proportional to the covariance matrix of $\mathbf{x}$, i.e. $W \propto \Sigma^{-1}$, $\Sigma := \mathrm{Cov}(\mathbf{x})$. When $W = I$, the loss reduces to empirical least square.

Next, we denote the privacy mechanism as an operator. Let $\mathcal{P}_\mu^{\mathcal{H}} : \mathcal{H} \to \widetilde{\mathcal{H}}$ denote a μ-GDP mechanism for releasing private function-valued summaries, and let $\mathcal{P}_\mu^{\mathcal{X}^n} : \mathcal{X}^n \to \mathcal{X}$ denote a μ-GDP mechanism acting directly on the observed dataset. The latter may involve performing operations such as clipping, averaging, and Gaussian perturbation before releasing a privatized summary. The particular constructions of these mechanisms will be developed in Section 4.

Under the previous framework, the two privacy-estimation pipelines considered in this paper can be written compactly as follows.

Fit-Then-Protect (FTP). Each individual trajectory is first converted into a function through $\mathcal{F}$, the estimated functions are averaged, and the resulting function is privatized:

$$\widetilde{f}_{\mathrm{FTP}} = \mathcal{P}_\mu^{\mathcal{H}} \left(\frac{1}{n} \sum_{i=1}^n \mathcal{F}_{\mathrm{FTP}}(\mathbf{x}_i) \right). \tag{4}$$

Protect-Then-Fit (PTF). Privacy is first enforced on the observed dataset, after which the function estimator is applied to the released summary:

$$\widetilde{f}_{\mathrm{PTF}} = \mathcal{F}_{\mathrm{PTF}}\left(\mathcal{P}_{\mu}^{\mathcal{X}^n}(X)\right). \tag{5}$$

Importantly, although both pipelines satisfy the same prescribed privacy guarantee after appropriate calibration, they differ fundamentally in the order in which smoothing and privatization are performed. The primary goal of this paper is to understand how this ordering affects statistical accuracy.

Remark 12 (Relation to classical smoothing orderings) *Before introducing differential privacy, it is instructive to consider the analogous problem without privacy perturbation. In classical FDA, discretely observed trajectories are often first represented as smooth functions before subsequent functional analysis (Ramsay and Silverman, 2005). Motivated by this workflow and by direct mean function estimation from pooled observations in noisy functional data analysis (Yao et al., 2005), we consider two possible orderings of smoothing and aggregation. Specifically, to estimate the sample mean function $\bar{f}$, one may either smooth each individual trajectory and then average, $\hat{f}_{\mathrm{STA}} = n^{-1}\sum_{i=1}^{n}\mathcal{F}(x_i)$, or first average the observed trajectories and then smooth, $\hat{f}_{\mathrm{ATS}} = \mathcal{F}\left(n^{-1}\sum_{i=1}^{n}x_i\right)$. We refer to these two procedures as* Smooth-Then-Average *(STA) and* Average-Then-Smooth *(ATS), respectively. When the estimation operator $\mathcal{F}$ is linear, the two estimators coincide. But generally they differ for nonlinear estimators, for example when the optimization problem in Equation* (2) *involves active constraints or other nonlinear regularization. In some settings considered later in this paper, the two estimators become identical in the absence of observation noise. Nevertheless, even in such cases the corresponding privacy-preserving pipelines FTP and PTF remain fundamentally different because the privacy mechanism itself is applied at different stages of the estimation procedure.*

Note that the two fitting operators are allowed to differ because the statistical characteristics of their inputs differ. In particular, the PTF estimator operates on privacy-perturbed observations whose covariance structure may vary across sampling locations. Accordingly, $\mathcal{F}_{\mathrm{PTF}}$ may employ a loss function tailored to the privacy-induced noise, such as weighted least squares. When the privatized observations have the same covariance structure across sampling locations, the two operators may coincide.

The preceding definitions describe FTP and PTF at the operator level. The actual implementation requires specifying the function space, fitting loss, fitting radius, privacy mechanism, and the finite-dimensional representation used for computation. We give Algorithms 1 and 2 for implementing the procedures.

Algoritm 1 and 2 provide concrete realization of the abstract operator $\mathcal{P}_{\mu}^{\mathcal{H}}$, $\mathcal{P}_{\mu}^{\mathcal{X}^n}$ introduced in Section 3.2. Other constructions are possible, provided they satisfy the same privacy guarantee. The clipping constant $\mathbf{c} = (c_1, \ldots, c_T)^{\top}$ for PTF depends on both the function space under consideration and the sampling location t_j. For example, in the Gaussian RKHS studied later, the reproducing property implies $|f(t)| \le \|f\|_{\mathcal{H}}\sqrt{K(t,t)}$, so that under the Gaussian kernel, where $K(t,t) = 1$, one may simply take $c_j = B'$. The polynomial and RKHS cases enter through the specification of $\mathcal{H}$ and $\mathcal{L}$, together with the corresponding sensitivity.

Algorithm 1 Fit-Then-Protect (FTP); $\texttt{FTP}(X, \mathbf{t}, \mathcal{H}, \mathcal{L}, B', \mu, C)$

Input: Observed trajectories $X = (\mathbf{x}_1, \ldots, \mathbf{x}_n)^\top$ at design points $0 \le t_1 < \cdots < t_T \le 1$; function space $\mathcal{H}$; loss function $\mathcal{L}$ and radius B'; privacy budget μ; Gaussian covariance operator C.
for $i = 1, \ldots, n$ **do**
  [Fitting] $\hat{f}_i^{(c)} \leftarrow \operatorname{argmin}_{f \in \mathcal{H}, \|f\|_{\mathcal{H}} \le B'} \mathcal{L}(f; \mathbf{x}_i)$.
end for
$\overline{\hat{f}^{(c)}} \leftarrow n^{-1} \sum_{i=1}^n \hat{f}_i^{(c)}$.
Compute $\Delta = 2B'/n$.
Sample Gaussian Process $Z \sim \mathcal{N}_{\mathcal{H}}(0, C)$.
[Protecting] $\widetilde{f}_{\mathrm{FTP}} \leftarrow \overline{\hat{f}^{(c)}} + \sigma Z, \sigma = \Delta/\mu$.
Output: $\widetilde{f}_{\mathrm{FTP}}$.

Algorithm 2 Protect-Then-Fit (PTF); $\texttt{PTF}(X, \mathbf{t}, \mathcal{H}, \mathcal{L}, B', \mu, \mathbf{c})$

Input: Observed trajectories $X = (\mathbf{x}_1, \ldots, \mathbf{x}_n)^\top$ at design points $0 \le t_1 < \cdots < t_T \le 1$; function space $\mathcal{H}$; loss function $\mathcal{L}$ and radius B'; privacy budget μ; clipping bound $c_j, j = 1, \ldots, T$.
for $j = 1, \ldots, T$ **do**
  Compute $\bar{x}_j^{(\mathrm{clip})} = n^{-1} \sum_{i=1}^n \mathrm{clip}_{[-c_j, c_j]}(x_{ij}), \quad \Delta_j = 2c_j/n, \quad \sigma_{\mathrm{DP},j} = \Delta_j \sqrt{T}/\mu$.
  Sample $Z_j \overset{i.i.d.}{\sim} N(0, \sigma_{DP,j}^2)$,
  [Protecting] Compute $\widetilde{x}_j = \bar{x}_j^{(\mathrm{clip})} + Z_j$
end for
[Fitting] $\widetilde{f}_{\mathrm{PTF}} \leftarrow \operatorname{argmin}_{f \in \mathcal{H}, \|f\|_{\mathcal{H}} \le B'} \mathcal{L}(f; \widetilde{x}), \quad \widetilde{x} = (\widetilde{x}_1, \ldots, \widetilde{x}_T)^\top$.
Output: $\widetilde{f}_{\mathrm{PTF}}$.

3.3 Utility Criteria

To compare the two procedures, we evaluate their estimation accuracy relative to the sample mean function $\bar{f}$. For finite-dimensional function spaces with $L^2[0,1]$ norm, we consider the integrated mean squared error (IMSE),

$$\mathrm{IMSE}(\widetilde{f}) = \int_{\mathcal{T}} \mathbb{E}\left[\widetilde{f}(t) - \bar{f}(t)\right]^2 dt. \tag{6}$$

For settings of infinite-dimensional function spaces, we instead consider the empirical mean squared error

$$\mathrm{MSE}(\widetilde{f}) = \frac{1}{T} \sum_{j=1}^{T} \mathbb{E}\left[\widetilde{f}(t_j) - \bar{f}(t_j)\right]^2. \tag{7}$$

Here, we use different metrics for different function spaces. For finite-dimensional function spaces, we can recover the underlying true function under certain conditions (e.g., in the absence of observation noise), making it possible to compare the global performance of the estimator with that of the true mean function. In infinite-dimensional spaces, however, there can be many different underlying functions that interpolate the same observed data

points, making the true function unidentifiable. Therefore, we can only consider metrics defined based on the observed data.

The remainder of the paper develops concrete realizations of the privacy operators in different function spaces and compares the statistical performance of the FTP and PTF pipelines under a common privacy budget. We write $\mathrm{IMSE}_{\mathrm{FTP}} = \mathrm{IMSE}(\widetilde{f}_{\mathrm{FTP}})$ for short and similarly in the following.

4 Theoretical Results

In this section, we develop the GDP mechanisms for functionals and compare the *FTP* and *PTF* regimes under the IMSE/MSE metric (depends on the function space). In Secion 4.1, we extend the GDP mechanisms for functionals to the general Banach space and apply it to the Polynomial space and RKHS. In Secion 4.2, we give results for IMSE/MSE Ratio between the two regimes under the case without observation noise.

4.1 Privacy Guarantees

We first establish the privacy guarantees for the two abstract privacy operators introduced in Section 3.

Theorem 13 (GDP for function-valued summaries) *Let $\mathbb{B}$ be a real separable Banach space, and let $S : \mathcal{X}^n \to \mathbb{B}$ be a deterministic summary operator. Let $\mathcal{H}$ be the corresponding Cameron–Martin space of a centered Gaussian Process $Z \sim \mathcal{N}_{\mathbb{B}}(0, C)$ with the covariance operator C. Assume that S is **compatible** with Z. Define the global sensitivity of S as $\Delta = \sup_{X \sim X'} \|S(X) - S(X')\|_{\mathcal{H}}$. Define*

$$\widetilde{S}(X) = S(X) + \sigma Z, \quad \sigma = \frac{\Delta}{\mu}.$$

Then $\widetilde{S}(X)$ satisfies μ-GDP over $\mathbb{B}$.

While Mirshani et al. (2019) extend previous results of Hall et al. (2013) from building (ϵ, δ)-DP in RKHS for point-wise evaluations of functions to Banach spaces and full-function releases, and Räisä et al. (2024) extends this to GDP under RKHS, Theorem 13 is the first to provide a generic GDP mechanism for function-valued summaries in the Banach space. Moreover, we provide a proof strategy directly works with the tradeoff function, which may have the potential to generalize to other types of DP, e.g. f-DP. Consequently, once the sensitivity of the functional statistic is bounded, the FTP pipeline in Equation (4) satisfies the prescribed μ-GDP guarantee simply by taking $S(X) = n^{-1} \sum_{i=1}^{n} \mathcal{F}(x_i)$, and calibrating the Gaussian process noise according to its global sensitivity. The remaining theoretical task is therefore to derive the global sensitivity Δ in the particular function spaces considered later.

Corollary 14 (Finite-dimensional realization) *Let $\phi_1, \ldots, \phi_m$ be orthonormal under $L^2[0,1]$ and consider $\mathcal{H} = \mathrm{span}\{\phi_1, \ldots, \phi_m\}$ with $L^2[0,1]$ norm, denoted as $\|\cdot\|_{\mathcal{H}}$. Any function $f \in \mathcal{H}$ can be represented as $f(t) = \theta^\top \phi(t) = \sum_{k=1}^{m} \theta_k \phi_k(t), t \in \mathcal{T}, \theta \in \mathbb{R}^m$, then we have*

$$\Delta^2 = \sup_{X \sim X'} \|f_X - f_{X'}\|_{\mathcal{H}}^2 = \sup_{X \sim X'} \|\theta_X - \theta_{X'}\|_2^2,$$

and $\widetilde{f}_X = (\theta_X + Z)^T\phi$ *with* $Z \sim N(0, \sigma^2 I_m)$ *achieves* μ*-GDP over* $\mathbb{B}$ *with* $\sigma = \dfrac{\Delta}{\mu}$.

Corollary 14 shows that the proposed functional GDP mechanism naturally extends the classical Gaussian mechanism from finite-dimensional vectors to function-valued summaries. Polynomial models therefore serve as a bridge between the familiar multivariate setting and the infinite-dimensional results developed later for RKHS. Moreover, we have to limit the number of basis elements in practice. Thus, Corollary 14 is prefered in implementation.

Proposition 15 (Privacy guarantee for PTF) *Under the functional data model given by Equation* (1) *and Assumption* A.1*, the estimator* $\widetilde{f}_{\mathrm{PTF}}$ *given by Algorithm 2 satisfies* μ*-GDP.*

Proof Since $\mathcal{F}_{\mathrm{PTF}} : \mathcal{X} \to \mathcal{H}$ is a deterministic measurable mapping, the result follows immediately from the post-processing property and that $\widetilde{x}$ satisfies μ-GDP (by composition). ■

4.2 Statistical Comparison

The privacy guarantees established in the previous subsection hold for a general Banach space. The statistical performance of the FTP and PTF pipelines, however, depends strongly on the structure of the underlying estimation problem. In this subsection we compare the two procedures in two representative settings.

We first study finite-dimensional polynomial spaces, where we use the IMSE for the comparison. We then turn to RKHS equipped with the Gaussian kernel, where the comparison is performed under the empirical MSE criterion.

For each function space, we first analyze the noiseless setting ($\sigma_0 = 0$), which reveals the intrinsic effect of the ordering of estimation and privatization. We then discuss how observation noise modifies the comparison and identify the additional bias introduced by nonlinear projection onto the feasible parameter space.

4.2.1 Polynomial spaces

Let $\phi_i : [0,1] \to \mathbb{R}$ for $i = 1, \ldots, m$ be the orthonormal basis with respect to $L^2[0,1]$ norm and inner product. We consider $\phi = (\phi_1, \ldots, \phi_m)^\top$ and denote the m-dimensional polynomial space as

$$\mathcal{H}^m_{\mathrm{poly}} = \left\{ \theta^\top \phi(t) : \theta \in \mathbb{R}^m, t \in [0,1] \right\},$$

and θ is the corresponding coefficient vector. Then the latent functional trajectories in $\mathcal{H}^m_{\mathrm{poly}}$ becomes $f_i(t) = \theta_i^\top \phi(t)$. Denote $\Phi = (\phi(t_1), \ldots, \phi(t_T))^\top \in \mathbb{R}^{T \times m}$, $M = \Phi^\top \Phi \in \mathbb{R}^{m \times m}$ with the following assumption

A.3 $M = \Phi^\top \Phi$ is positive definite.

Assumption **A.3** depends on both the choice of orthonormal basis $\phi(t)$ and the evaluation points $t_1, \ldots, t_T$. Equivalently, it requires the evaluation matrix Φ to have full column rank m. In particular, $T \geq m$ is necessary. If the basis $\phi(t)$ form an orthonormal basis

of the polynomial space of degree at most $m-1$, then Assumption **A.3** holds if and only if the evaluation points $\mathbf{t}$ contain at least m distinct values. For a general basis $\phi(t)$, the corresponding condition on the evaluation points may differ.

Definition 16 (Spectral condition number) *Let $A \in \mathbb{R}^{n\times n}$ be a square matrix, a scalar λ_i is an eigenvalue of A if there exists a nonzero vector $\mathbf{v}_i$ such that $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$ for $i = 1, \ldots, n$. For a symmetric positive-definite matrix A, the spectral condition number is defined as $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$, where $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ denote the smallest and largest eigenvalues of A, respectively.*

The spectral condition number quantifies the conditioning of a symmetric positive-definite matrix by measuring the ratio between its largest and smallest eigenvalues. A condition number close to 1 indicates that the eigenvalues are relatively uniform, whereas a large condition number indicates greater variation in the eigenvalues and potentially greater sensitivity to perturbations. The spectral condition number can be useful for characterizing the geometry and numerical stability of covariance matrices, as well as understanding how the eigenvalue distribution of a matrix influences the performance of matrix-based mechanisms and associated error bounds.

Theorem 17 (FTP IMSE upper bound in polynomial spaces) *Under the polynomial space $\mathcal{H}^m_{\text{poly}}$ with Assumptions A.1, A.2, and A.3. For the estimator $\widetilde{f}_{\text{FTP}}$ given by $\texttt{FTP}(X, \mathbf{t}, \mathcal{H}, \mathcal{L}, B', \mu, C)$ (Algorithm 1) with $\mathcal{L}$ defined in Equation (3) and $W = I$. When $B' \geq B + \dfrac{\sqrt{m}\sigma_0}{\sqrt{\lambda_{\min}(M)}}$, $\widetilde{f}_{\text{FTP}}$ satisfies*

$$\begin{aligned}\text{IMSE}_{\text{FTP}} \leq & m \cdot \frac{4B'^2}{n^2\mu^2} + \frac{\kappa(M)\sigma_0^2}{n}\operatorname{tr}(M^{-1}) \\ & + \left(1 - \frac{1}{n}\right)\sqrt{3}\kappa(M)\sigma_0^2 \operatorname{tr}(M^{-1}) \exp\left[-\frac{1}{4}\left(\frac{(B'-B)\sqrt{\lambda_{\min}(M)}}{\sigma_0} - \sqrt{m}\right)^2\right].\end{aligned} \tag{8}$$

Theorem 17 gives the upper bound for IMSE of FTP under polynomial space, which is determined by the DP mechanism, observation noise, and the active constraint effect (similar to clipping effect). When $\sigma_0 = 0$, the equality holds. Furthermore, if there exist some constants $C_1 > 2$ and $C_2 > 0$ such that

$$B + \frac{\sigma_0}{\sqrt{\lambda_{\min}(M)}}(\sqrt{m} + C_1\sqrt{\log n}) \leq B' \leq C_2\sqrt{n},$$

we have $\text{IMSE}_{\text{FTP}} = O\left(\dfrac{1}{n}\right)$. Notice that as $T \to \infty$, $\dfrac{1}{T}M \to I_m$ and $\operatorname{tr}(M^{-1}) \to \dfrac{m}{T}$.

Theorem 18 (PTF IMSE lower bound in polynomial spaces) *Under the polynomial space $\mathcal{H}^m_{\text{poly}}$ with Assumptions A.1, A.2 with $\sigma_0 = 0$, and A.3. Denote $\tau = \dfrac{2B'\sqrt{T}}{n\mu}$. For*

the estimator $\widetilde{f}_{\mathrm{PTF}}$ *given by* $\mathtt{PTF}(X, \mathbf{t}, \mathcal{H}, \mathcal{L}, B', \mu, \mathbf{c})$ *(Algorithm 2) with* $\mathcal{L}$ *defined in Equation* (3)*,* $W = \tau^2\Sigma^{-1}$*,* $\Sigma = \mathrm{Cov}(\widetilde{x})$*,* $c_j = B'\|\phi(t_j)\|_2$*. Denote* $A = \Phi^\top W \Phi$*, then* A *is positive definite. When* $B' \geq B + \dfrac{\tau\sqrt{m}}{\sqrt{\lambda_{\min}(A)}}$*,* $\widetilde{f}_{\mathrm{PTF}}$ *satisfies*

$$\mathrm{IMSE}_{\mathrm{PTF}} \geq \tau^2 \operatorname{tr}(A^{-1}) \left\{ 1 - \sqrt{3} \exp\left[-\frac{1}{4} \left\{ \frac{(B'-B)\sqrt{\lambda_{\min}(A)}}{\tau} - \sqrt{m} \right\}^2 \right] \right\}. \tag{9}$$

Theorem 18 gives the lower bound for IMSE of PTF under the polynomial space without observation noise, when $B' \geq B + \dfrac{\tau}{\sqrt{\lambda_{\min}(A)}}(\sqrt{m} + C\sqrt{\log n})$, the exponential term decays rapidly, which leads to our following corollary.

Corollary 19 (IMSE Ratio for Polynomials) *Assume* $\sigma_0 = 0$*, and the FTP and PTF estimators defined in Theorem 17 and Theorem 18, denote the IMSE ratio of FTP to PTF as* $R_1(n, B', m, T) = \dfrac{\mathrm{IMSE}_{\mathrm{FTP}}}{\mathrm{IMSE}_{\mathrm{PTF}}}$*. Then, for any fixed* T*, when* $B' \in \Omega(\sqrt{\log n})$*,*

$$\lim_{n\to\infty} R_1(n, B', m, T) \leq \frac{1}{m}. \tag{10}$$

Corollary 19 suggests that the relative advantage of the FTP regime can increase with the effective dimension m, defined as the number of basis functions. Here, m is distinct from the polynomial degree. For example, when $m = 2$, one may consider the orthonormal basis $\{1, \sqrt{3}(2t-1)\}$ for $\mathrm{span}\{1, t\}$, or the orthonormal basis $\{1, \sqrt{5}(3t^2-1)/2\}$ for $\mathrm{span}\{1, t^2\}$. Both choices satisfy the same upper bound $R_1 \leq 1/m = 1/2$, provided the assumptions of the corollary hold.

4.2.2 RKHS

Let $\mathcal{H}_K$ be the RKHS associated with kernel $K(\cdot,\cdot)$, denote $K_T = [K(t_k, t_l)]_{k,l=1}^T$ with the assumption

A.4 K_T is positive definite.

Theorem 20 (MSE upper bound for FTP in RKHS) *Under the RKHS* $\mathcal{H}_K$ *with Assumptions A.1, A.2, and A.4. For the estimator* $\widetilde{f}_{\mathrm{FTP}}$ *given by* $\mathtt{FTP}(X, \mathbf{t}, \mathcal{H}, \mathcal{L}, B', \mu, C)$ *(Algorithm 1) with* $\mathcal{L}$ *defined in Equation* (3) *and* $W = I$*. When* $B' \geq B + \dfrac{\sigma_0\sqrt{T}}{\sqrt{\lambda_{\min}(K_T)}}$*, we have*

$$\begin{aligned}\mathrm{MSE}_{\mathrm{FTP}} \leq{}& \frac{4B'^2}{n^2\mu^2 T} \operatorname{tr}(K_T) + \frac{\sigma_0^2}{n} \\ &+ \left(1 - \frac{1}{n}\right) \sigma_0^2 \sqrt{\frac{T+2}{T}} \exp\left\{ -\frac{1}{4} \left(\frac{(B'-B)\sqrt{\lambda_{\min}(K_T)}}{\sigma_0} - \sqrt{T} \right)^2 \right\}.\end{aligned}$$

Theorem 20 provides a general upper bound for the MSE of FTP in the RKHS setting. The result is analogous to Theorem 17, but the present formulation allows the true functions f_i to belong to the full, potentially infinite-dimensional RKHS. The bound consists of three components: the privacy error, the observation-noise error, and the error induced by the active RKHS-norm constraint. Under fixed T, suppose that

$$B + \frac{\sigma_0}{\sqrt{\lambda_{\min}(K_T)}}\left(\sqrt{T} + C_1\sqrt{\log n}\right) \le B' \le C_2\sqrt{n}$$

for some constants $C_1, C_2 > 0$. Then the clipping term decays polynomially in n. In particular, when $C_1 > 2$, the clipping term is $o(1/n)$, and therefore the dominant terms in the MSE bound are the privacy and observation-noise terms, both of which are $O(1/n)$. Consequently,

$$\mathrm{MSE}_{\mathrm{FTP}} = O\left(\frac{1}{n}\right).$$

For the Gaussian kernel, $K(t_j, t_j) = 1$ for every j, and hence $\mathrm{tr}(K_T) = T$. Therefore, the privacy contribution reduces to $\dfrac{4B'^2}{n^2\mu^2}$. Thus, if $B' = O(\sqrt{n})$, the privacy error is also $O(1/n)$. This shows that, under fixed T, the estimator achieves the standard $O(1/n)$ MSE rate provided that the constraint radius B' grows sufficiently slowly to control the privacy error while sufficiently quickly to make the probability of an active constraint negligible.

Theorem 21 (MSE lower bound for PTF in RKHS) *Under the RKHS $\mathcal{H}_K$ with Assumptions A.1, A.2 with $\sigma_0 = 0$, and A.4. Denote $\tau = \dfrac{2B'\sqrt{T}}{n\mu}$. For the estimator $\widetilde{f}_{\mathrm{PTF}}$ given by $\mathtt{PTF}(X, \mathbf{t}, \mathcal{H}, \mathcal{L}, B', \mu, \mathbf{c})$ (Algorithm 2) with $\mathcal{L}$ defined in Equation (3), $W = \tau^2\Sigma^{-1}$, $\Sigma = \mathrm{Cov}(\widetilde{x})$, $c_j = B'\sqrt{K(t_j, t_j)}$. Denote $A = K_T^{1/2} W K_T^{1/2}$, then A is positive definite. When $B' - B \ge \dfrac{\tau\sqrt{T}}{\sqrt{\lambda_{\min}(A)}}$, we have*

$$\mathrm{MSE}_{\mathrm{PTF}} \ge \frac{\tau^2}{T}\,\mathrm{tr}(W^{-1})\left[1 - \sqrt{3}\exp\left\{-\frac{1}{4}\left(\frac{(B'-B)\sqrt{\lambda_{\min}(A)}}{\tau} - \sqrt{T}\right)^2\right\}\right]. \tag{11}$$

Theorem 21 provides a lower bound for MSE of PTF under RKHS without observation noise, which is similar to Theorem 18. In particular, under the Gaussian Kernel, we have $K(t_j, t_j) = 1, \forall j$, therefore $W^{-1} = \mathrm{diag}\{K(t_j, t_j)\}$ and $\mathrm{tr}(W^{-1}) = T$.

Corollary 22 (MSE Comparison under RKHS) *Assume $\sigma_0 = 0$, and the FTP and PTF estimators defined in Theorem 20 and Theorem 21, denote the MSE ratio of FTP to PTF as $R_2(n, B', m, T) = \dfrac{\mathrm{MSE}_{\mathrm{FTP}}}{\mathrm{MSE}_{\mathrm{PTF}}}$. Then, for any fixed T, when $B' \in \Omega(\sqrt{\log n})$,*

$$\lim_{n\to\infty} R_2(n, B', m, T) \le \frac{1}{T}. \tag{12}$$

Corollary 22 shows that for $T > 1$ under noiseless case, FTP achieves strictly smaller empirical MSE than PTF, with an improvement factor of T.

5 Simulation Studies

In this section, we empirically implement our proposed strategies of *Protect-Then-Fit* and *Fit-Then-Protect* via Algorithm 1 and Algorithm 2 in the selected polynomial space, and RKHS with Gaussian kernel. In Section 5.1, we first validate our theoretical upper bounds in Section 4.2, and in Section 5.2, we show how different choices of the constraint radius B' affect the performance of FTP and PTF. Finally, in Section 5.3, we further explore simulations with different magnitudes of observation noise and discuss its effect. All the simulations are run with 5000 replicates. We specify our settings in the two function spaces as below.

Polynomial space. Consider the functional model $f_1, \ldots, f_n \in \mathcal{H}_m$ with $L^2[0,1]$ norm and inner product, where $\mathcal{H}_m = \left\{\theta^\top \phi(t) : \theta \in \mathbb{R}^m, t \in [0,1]\right\}$, and $\phi(t) = (\phi_1(t), \ldots, \phi_m(t))^\top$ is the orthonormal basis satisfying that the polynomial degree of $\phi_k(t)$ is $k-1$ for $k = 1, \ldots, m$. For example, when $m = 2$, the orthonormal basis, is $(\phi_1(t), \phi_2(t))^\top = (1, \sqrt{12}(t - 0.5))^\top$. We construct this basis iteratively based on *shifted orthonormal Legendre basis*. Then, for each i, we generate $f_i(t) = \theta_i^\top \phi(t)$ with θ_i i.i.d. in the m-dimensional ball with radius $B = 1$, which ensures $\|f_i\|_{L^2} = \|\theta_i\|_2 \leq B$. The data matrix X is generated based on the model: $x_{ij} = \theta_i^\top \phi(t) + \epsilon_{ij}, \epsilon_{ij} \overset{i.i.d.}{\sim} N(0, \sigma_0^2)$, with an equally spaced location grid between 0 and 1. Specifically, $t_j = (j-1)/(T-1), j = 1, \ldots, T$ and we fix $T = 20$. Then we obtain the two private estimators $\widetilde{f}_{\mathrm{FTP}}$ through Algorithm 1 and Algorithm 2, Finally, we use the IMSE metric to evaluate the two estimators compared to the true sample mean function $\bar{f} = n^{-1} \sum_{i=1}^n f_i$. We try different settings for sample sizes, basis dimensions, privacy budgets, constraint radius, and observation noise variances, where $n \in \left\{2^2, \ldots, 2^{11}\right\}$, $m \in \{2, 4, 6, 8\}$, $\mu \in \{0.1, 1\}$, $B' \in \left\{B, \sqrt{\log n}, \log n, \sqrt{n}, n^{2/3}\right\}$, and $\sigma_0 \in \{0, 1, 100\}$, respectively.

RKHS. Consider the RKHS induced by the Gaussian (RBF) kernel

$$K(s,t) = \exp\left\{-\frac{(s-t)^2}{2\rho^2}\right\}, \ s, t \in [0,1]$$

with bandwidth $\rho = 0.2$. The functions are evaluated on an equally spaced grid on $[0,1]$ with $T = 50$ points, and are generated from $M = 20$ kernel centers $\{c_1, \cdots, c_M\}$. For each user, we generate an RKHS function whose RKHS norm is bounded by $B = 1$ by sampling its coefficient vector α_i uniformly within the corresponding B-bounded ball. We use a nonzero population mean direction in generating the coefficients so that the underlying mean function is nontrivial. The clean functional observations are then generated by evaluating these functions on the observation grid, and the observed data are obtained according to

$$x_{ij} = f_i(t_j) + \epsilon_{ij}, \qquad \epsilon_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_0^2).$$

$$f_i = \alpha_i^\top K(t), K(t) = (K(c_1, t), \ldots, K(c_M, t))^\top.$$

For the privacy mechanisms, we use $B' = 1$ as the assumed RKHS norm bound and consider $n \in \{50, 100, \ldots, 500\}$ and $\mu \in \{0.1, 0.2, \ldots, 1\}$. For each fixed (n, μ, σ_0) setting, the same generated dataset is used across 25,000 repetitions of the privacy mechanisms, with independent privacy noise added in each repetition. The observation-noise levels considered are $\sigma_0 \in \{0.01, 1, 100\}$.

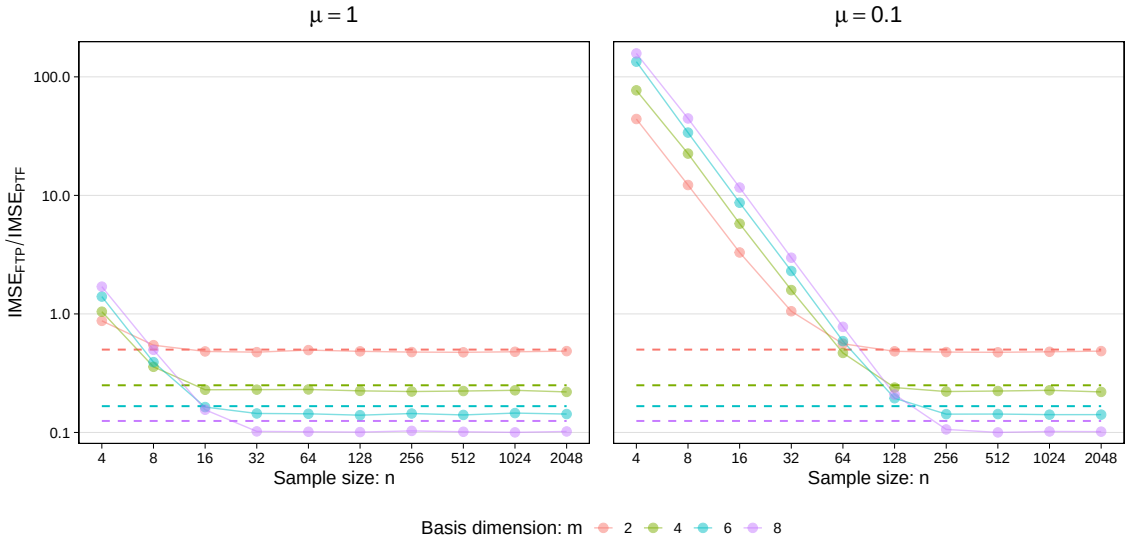


Figure 1: Empirical IMSE ratio of FTP to PTF vs sample size n in polynomial space. Each solid curve is corresponding to a basis dimension m, and each dashed horizontal line represents the asymptotic upper bound of $1/m$.

5.1 Effect of Effective Basis Dimension

In Figure 1, we plot the IMSE ratio of FTP to PTF across different sample size n and basis dimension $m \in \{2, 4, 6, 8\}$, for privacy budget $\mu = 0.1, 1$ under noiseless case ($\sigma_0 = 0$). Under each setting given m, it shows that the IMSE ratio decreases as n increases. For a sufficient large n, each curve stabilizes below the reference dashed line of $1/m$, indicating that our upper bound result in Corollary 19 is valid, and it suggests that FTP will outperform PTF asymptotically when $m \geq 2$. Moreover, one can observe that for larger m and smaller μ, the required sample size n for achieving desiared IMSE ratio upper bound becomes larger. As both settings will lead to a larger DP noise variance and less data utility, and makes the constraint for PTF active with high probability under same n (add Table in Appendix). Figure 2 shows all these similar patterns in RKHS, though the required sample size for the MSE ratio of FTP to PTF to fall below corresponding upper bound is larger than the polynomial case under same effective basis dimension $T = m$. For example, when $m = T = 8$ with $\mu = 1$, the required sample size in polynomial space is 32 where as in RKHS it is 256.

5.2 Choices of Constraint Radius

In Figure 3, we plot the scaled IMSE (multiply by n) of FTP, PTF, and the ratio of FTP to PTF across different sample size n and constraint radius B'. The IMSE is scaled by n to help capture the its ratio, for example, when $B' = \sqrt{n}$, the corresponding curve is almost horizontal, suggesting that the IMSE for both FTP and PTF is $\Omega(1/n)$ under this choice

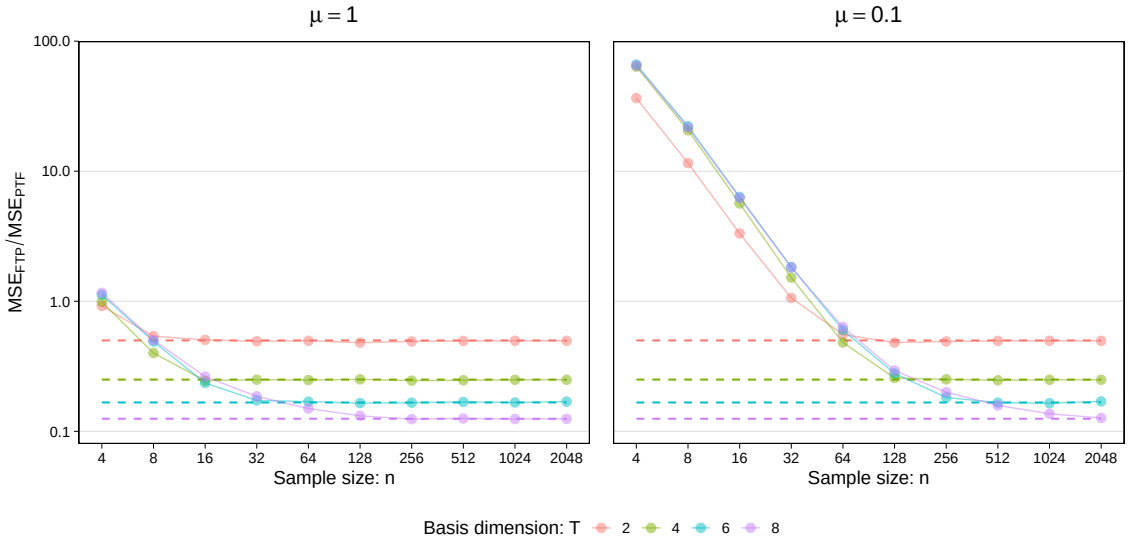


Figure 2: Log-log Empirical MSE ratio of FTP to PTF vs sample size n in RKHS. Each solid curve is corresponding to an effective basis dimension T, and each dashed horizontal line represents the theoretical upper bound of $1/T$.

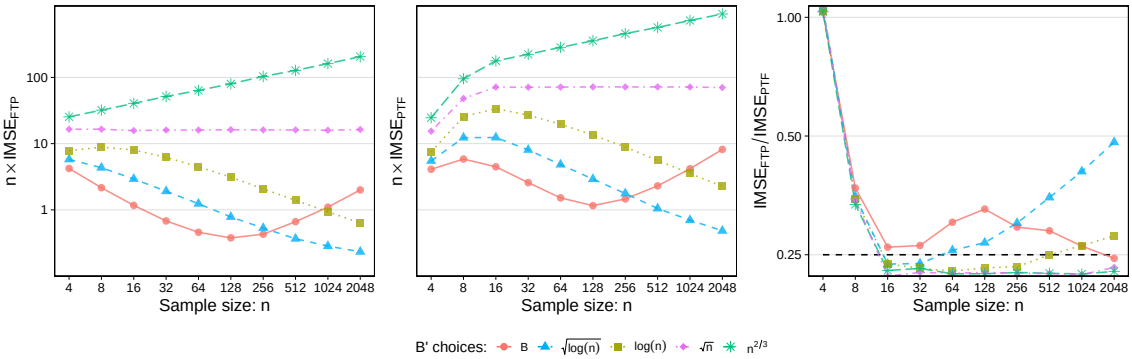


Figure 3: Log-log plot of empirical IMSE×n of FTP, PTF and the ratio vs sample size n in polynomial space with different B' choices.

of constraint radius. We include $B' = B$ as the oracle baseline as it represents the true knowledge of data. We choose $\sqrt{\log n}$, $\log n$ and $\sqrt{n}$ as three representative rates based on our theoretical analysis in Section 4.2, and $B' = n^{2/3}$ is out of the conditions for $O(1/n)$ IMSE rate of FTP. Figure 3 shows that true B is the best choice when sample size is small, while $\sqrt{\log n}$ and $\log n$ seem to be the better choices when n is sufficiently large, and could

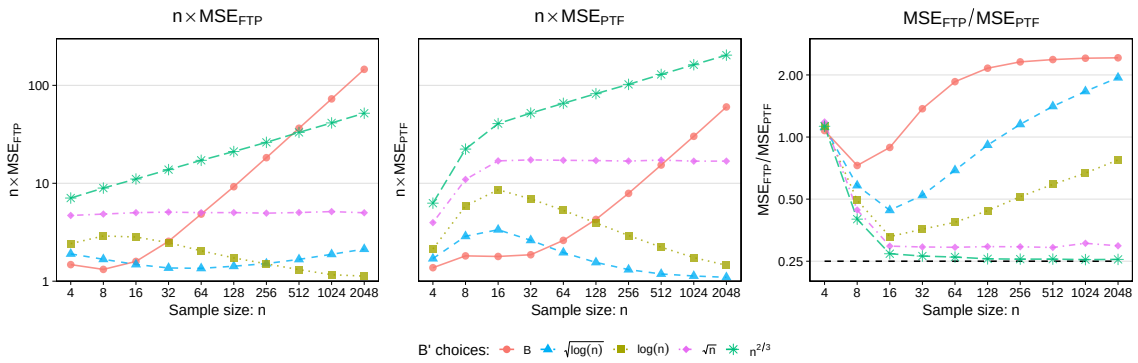


Figure 4: Log-log plot of empirical MSE×n of FTP, PTF and the ratio vs sample size n in RKHS with different B' choices.

achieve $o(1/n)$ IMSE rate for both FTP and PTF. However, neither of the two choices could achieve the same ratio bound as the in noiseless setting. That says, as n increases, the relative advantage of FTP to PTF will be weaker. Figure 4 shows similar results, while the MSE ratio is not always bounded by 1, which suggests the complicated interaction of observation noisy and leads to our discussion in the next point.

5.3 Effect of Observation Noise

We further explore the effect of observation noise compared to noiseless case. From Figure 5, we can observe that when observation noise is small ($\sigma_0 = 0.01$), the IMSE (FTP/PTF) ratio behavior is still similar to the noiseless setting, as there is not an increasing trend across n or μ and the curves are centered around the theoretical ratio for. When σ_0 increases to 1, the empirical curves are mostly above the dash line, and exhibits an obvious increasing trend over n, μ, although seems still bounded below 1. As σ_0 becomes larger to 100, the ratio eventually moves beyond 1, which vanishes the advantages of FTP.

6 Discussion and Directions for Future Work

In this paper, we studied differential privacy for functional mean estimation from the perspective of the ordering between functional estimation and privacy perturbation. We introduced the Fit-Then-Protect (FTP) and Protect-Then-Fit (PTF) pipelines and developed a common GDP framework for comparing their statistical efficiency. Our theoretical results show that, under the noiseless settings considered, the ordering of estimation and privatization can substantially affect estimation accuracy even when both procedures satisfy the same privacy guarantee. In particular, FTP achieves smaller estimation error than PTF in the Polynomial and RKHS settings studied here, with the asymptotic error ratio governed by the effective dimension in the polynomial case and by the number of sampling locations in the RKHS case. These results highlight the importance of exploiting functional structure before privatization: FTP calibrates privacy noise to an aggregated functional

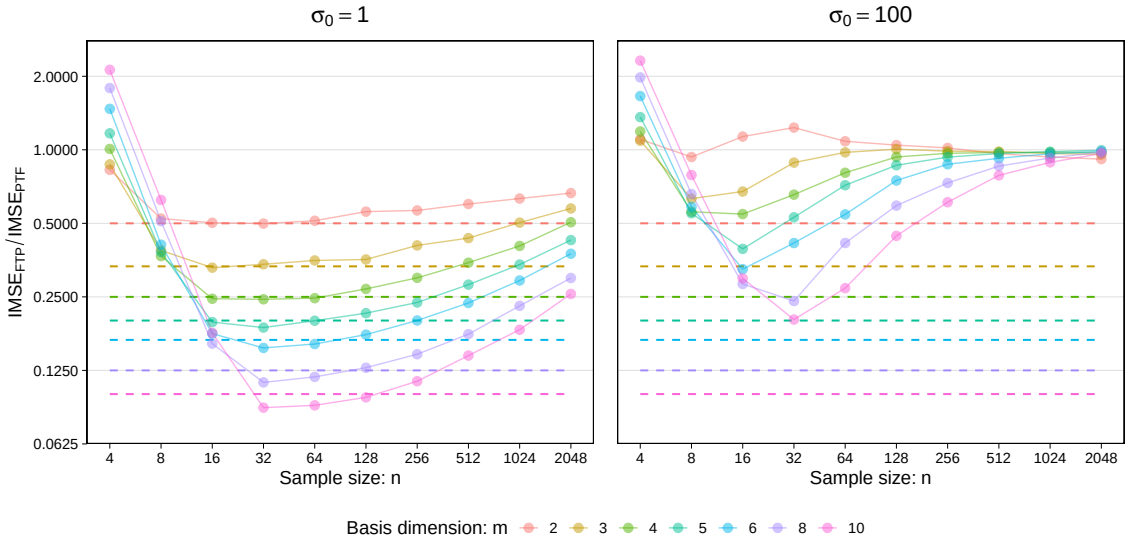


Figure 5: Log-log plot of empirical IMSE ratio of FTP to PTF vs sample size n in polynomial space with $\sigma_0 = 1$ and 100. Each solid curve is corresponding to a basis dimension m, and each dashed horizontal line represents the theoretical upper bound of $1/m$.

representation, whereas PTF introduces privacy perturbations at the observation level and subsequently propagates them through the functional estimation procedure. More broadly, the proposed Banach-space GDP framework shows that the choice of function-space geometry and Gaussian perturbation can be incorporated directly into the design of private functional estimators.

The simulation results further demonstrate that the relative performance of FTP and PTF depends on the observation regime. In the noiseless and low-noise settings, the empirical results are consistent with the theoretical comparisons and show a substantial advantage for FTP. As observation noise increases, however, this advantage becomes smaller and can eventually disappear or reverse. This behavior emphasizes that the benefit of applying functional structure before privatization is not independent of the underlying statistical noise. Rather, the relative performance reflects the interaction between privacy noise, observation noise, function-space geometry, and estimator complexity. The constraint radius B' also plays an important role in this trade-off: increasing B' reduces the probability of an active constraint but simultaneously increases the magnitude of the privacy noise. Thus, sensitivity calibration and statistical estimation must be considered jointly when designing private functional procedures.

Several limitations of the current analysis suggest directions for future work. First, our direct theoretical comparison between FTP and PTF focuses primarily on the noiseless setting. Although we derive noisy-data bounds for FTP and investigate observation

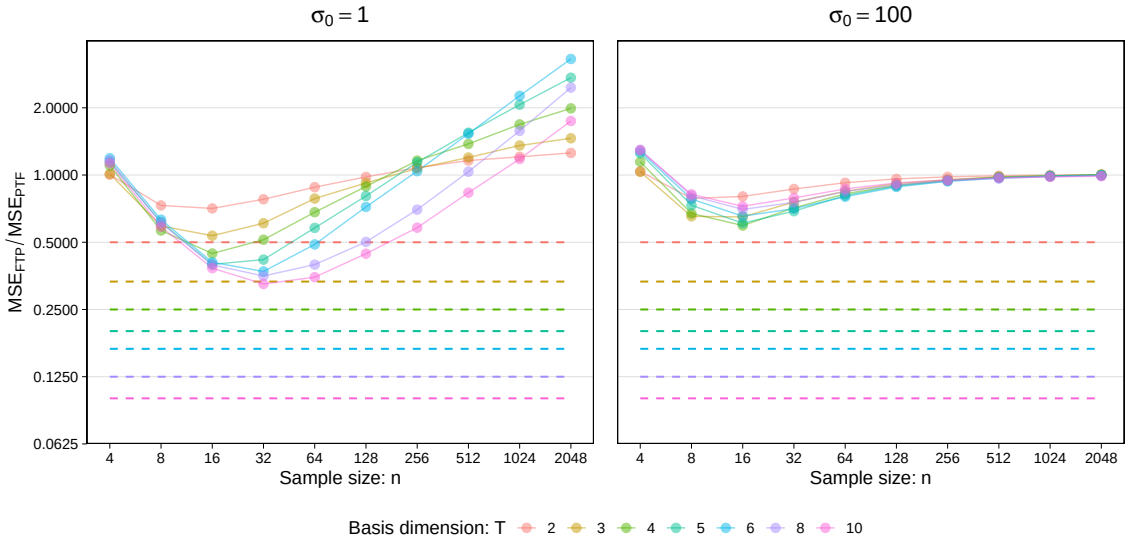


Figure 6: Log-log plot of empirical MSE ratio of FTP to PTF vs sample size n in RKHS with $\sigma_0 = 1$ and 100. Each solid curve is corresponding to a basis dimension T, and each dashed horizontal line represents the theoretical upper bound of $1/T$.

noise empirically, a general theoretical characterization of the FTP–PTF comparison with nonzero observation noise remains open. Such a result would need to account simultaneously for observation-noise covariance, privacy-induced covariance, and the nonlinear effects of constrained functional estimation. A more complete noisy-data theory could clarify how the relative efficiency of the two orderings changes across observation regimes and identify conditions under which their performance becomes comparable or one procedure may become preferable.

Second, the current framework assumes a common observation grid and considers polynomial spaces and Gaussian-kernel RKHSs as representative examples. Extending the analysis to irregularly sampled functional data and to broader classes of function spaces, such as Sobolev spaces and RKHSs with different eigenvalue decay properties, would help characterize how sampling design, smoothness, and effective dimension influence the privacy-utility trade-off. Such extensions may also require covariance structures for the Gaussian perturbation that are adapted to the underlying function space rather than to the particular choices considered here. In addition, while the current PTF construction provides a convenient GDP guarantee through perturbation at the observation level, coordinatewise perturbation need not be optimal for the geometry of functional data. Characterizing the optimal GDP design for PTF, including structured perturbations that exploit basis, kernel, or covariance information, is therefore an important open problem. Such mechanisms could potentially improve the utility of PTF and provide a more geometry-aware comparison with FTP.

Finally, the FTP/PTF distinction extends naturally beyond functional mean estimation. Functional regression, functional principal component analysis, covariance-function estimation, and other functional data analysis procedures involve transformations of functional observations that may be performed either before or after privacy perturbation. Investigating how privacy-estimation ordering interacts with these procedures, particularly when the estimation operator is nonlinear, could lead to a broader theory of when statistical structure should be extracted before privatization and when privacy should instead be imposed at an earlier stage of the analysis.

Overall, our results suggest that differential privacy for functional data is not only a problem of calibrating an appropriate noise process, but also a problem of deciding where privacy should be imposed within the statistical pipeline. The comparison between FTP and PTF illustrates that this design choice can have a substantial effect on statistical efficiency, while also showing that its impact depends on the observation-noise regime and the geometry of the underlying function space. These findings motivate further development of privacy mechanisms that jointly account for functional structure, observation noise, sampling design, and the geometry of the statistical representation.

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Appendix A. Limitation of pure differential privacy

Definition 23 ((ϵ, δ)-Differential Privacy) *A randomized mechanism $M : \mathcal{D} \to \mathcal{Y}$ satisfies (ϵ, δ)-DP if for all adjacent datasets $D \sim D'$ that differ in one individual record and all measurable sets $A \subseteq \mathcal{Y}$,*

$$\Pr(M(D) \in A) \leq e^{\epsilon} \Pr(M(D') \in A) + \delta.$$

Here, ϵ controls the privacy loss; δ is a small probability of failure. When $\delta = 0$, the mechanism satisfies *pure* ϵ-DP. Intuitively, small values of ϵ and δ imply stronger privacy protection.

Proposition 24 (Laplace Mechanism, Dwork et al., 2006) *Given a statistic $S(D) \in \mathbb{R}^d$, the Laplace mechanism releases*

$$M(D) = S(D) + L,$$

where

$$L_i \overset{iid}{\sim} Lap\left(\frac{\Delta_1(S)}{\epsilon}\right).$$

The Laplace mechanism satisfies ϵ-DP.

In infinite-dimensional settings, extending Laplace-type mechanisms becomes difficult. In particular, some work shows that there does not exist a natural infinite-dimensional analogue of elliptical Laplace distributions that achieves pure ϵ-DP (Reimherr and Awan, 2019).

Example 2 (A motivating constant-function example) *To illustrate why the ordering of estimation and privacy may matter, consider the simplified setting in which there is no observation noise ($\sigma_0 = 0$) and each individual function is constant,*

$$f_i(t) \equiv c_i, \qquad |c_i| \leq B.$$

The target parameter is the sample mean

$$\bar{c} = \frac{1}{n} \sum_{i=1}^{n} c_i.$$

Under the FTP strategy, each constant function is first estimated exactly, the estimates are averaged, and a single privacy mechanism is applied to the resulting scalar mean. Since the L_1-sensitivity of the sample mean is $2B/n$, the Laplace mechanism releases

$$\widetilde{c}_{\mathrm{FTP}} = \bar{c} + \eta, \qquad \eta \sim \mathrm{Lap}\left(\frac{2B}{n\epsilon}\right),$$

which satisfies pure ϵ-DP.

By contrast, a natural implementation of PTF is to first compute the sample mean at each observation time,

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^{n} x_{ij},$$

privatize every coordinate independently, and then estimate the constant function from the released observations. Allocating the privacy budget equally across the T coordinates gives

$$\widetilde{x}_j = \bar{x}_j + \eta_j, \qquad \eta_j \overset{i.i.d.}{\sim} \mathrm{Lap}\left(\frac{2BT}{n\epsilon}\right),$$

where the factor T follows from the sequential composition theorem. Averaging the privatized coordinates yields a variance that is exactly T times larger than that of the FTP estimator.

In contrast, under μ-Gaussian Differential Privacy, the composition rule is quadratic rather than linear. If each coordinate is released under $\mu/\sqrt{T}$-GDP, then averaging the resulting Gaussian perturbations exactly offsets the $\sqrt{T}$ increase in the noise standard deviation, so both FTP and this coordinate-wise PTF construction achieve the same privacy-induced variance order. This toy example illustrates that the relative statistical performance of FTP and PTF depends not only on the estimation problem, but also on the privacy framework itself.

Appendix B. Proofs and Technical Details

Theorem 25 (Cameron–Martin Theorem) *Let $Z \sim \mathcal{N}_{\mathbb{B}}(0, C)$ and let $\mathcal{H}_C$ be its Cameron–Martin space. For any $h \in \mathcal{H}_C$, the laws of Z and $Z+h$ are mutually absolutely continuous. In contrast, if $h \notin \mathcal{H}_C$, the corresponding translated Gaussian measures are singular.*

Consequently, if Q denotes the law of σZ and $h \in \mathcal{H}_C$, then the law of $\sigma Z + h$ is absolutely continuous with respect to Q, and its Radon–Nikodym derivative has the Gaussian-shift form

$$\frac{dQ_h}{dQ}(z) = \exp\left\{\frac{1}{\sigma}\langle h, z\rangle_{\mathcal{H}_C} - \frac{1}{2\sigma^2}\|h\|^2_{\mathcal{H}_C}\right\},$$

where the first term is understood through the canonical Gaussian Cameron–Martin pairing.

Lemma 26 (Cameron-Martin Lemma) *Let $\mathbb{B}$ be a real separable Banach space, and let $S : \mathcal{X}^n \to \mathbb{B}$ be a deterministic summary operator. Let $\mathcal{H}$ be the corresponding Cameron–Martin space of a centered Gaussian Process $Z \sim \mathcal{N}_{\mathbb{B}}(0, C)$ with the covariance operator C. Assume that S is **compatible** with Z, i.e., $S(X)$ resides in $\mathcal{H}$ for every $X \in \mathcal{X}^n$. Denote Q as the probability measure over $\mathbb{B}$ induced by σZ, and $\{P_X : X \in \mathcal{X}^n\}$ the family of probability measures over $\mathbb{B}$ induced by $S(X) + \sigma Z$, then every measure P_X has a density over $\mathbb{B}$ with respect to Q, which is given by*

$$\frac{dP_X}{dQ}(h) = \exp\left\{-\frac{1}{2\sigma^2}\left(\|S(X)\|^2_{\mathcal{H}} - 2\langle S(X), h\rangle_{\mathcal{H}}\right)\right\},$$

where $\langle\cdot,\cdot\rangle_{\mathcal{H}}$ denotes the inner product of $\mathcal{H}$.

Proof [Proof for Theorem 13] Denote Q as the probability measure induced by σZ, P_X and $P_{X'}$ as corresponding probability measures based on $S(X) + \sigma Z$ and $S(X') + \sigma Z$, our goal is to compare the following two measures/distributions:

$$H_0 : \mathcal{N}_{\mathbb{B}}(S(X), \sigma^2 C) \qquad vs \qquad H_a : \mathcal{N}_{\mathbb{B}}(S(X'), \sigma^2 C)$$

Denote $T(P_D, P_{D'})$ as the tradeoff function between the two distributions, we apply **Neyman-Pearson lemma** (in the function space) to find the MP test, which depends on the likelihood ratio and can be defined based on Cameron-Martin Theorem and Radon-Nikodym derivatives

$$\frac{dP_X}{dQ}(h) = \exp\left\{-\frac{1}{2\sigma^2}\left(\|S(X)\|_{\mathcal{H}}^2 - 2\langle S(X), h\rangle_{\mathcal{H}}\right)\right\}$$
$$\frac{dP_{X'}}{dQ}(h) = \exp\left\{-\frac{1}{2\sigma^2}\left(\|S(X')\|_{\mathcal{H}}^2 - 2\langle S(X'), h\rangle_{\mathcal{H}}\right)\right\}$$

Denote the likelihood ratio as

$$\Lambda(x) = \frac{dP_{X'}}{dP_X}(h) = \frac{dP_{X'}}{dQ}(h)/\frac{dP_X}{dQ}(h)$$

Apply the previous results, we get

$$\begin{aligned}\Lambda(h) =& -\frac{1}{2\sigma^2}\left(\|S(X')\|_{\mathcal{H}}^2 - \|S(X)\|_{\mathcal{H}}^2 - 2\langle S(X') - S(X), h\rangle_{\mathcal{H}}\right)\\ =& -\frac{1}{2\sigma^2}\left(\|S(X) - S(X')\|_{\mathcal{H}}^2 - 2\langle S(X') - S(X), h - S(X)\rangle_{\mathcal{H}}\right)\end{aligned}$$

By the Neyman-Pearson Lemma, the optimal test is given by

$$\phi(h) = \mathbf{1}(\Lambda(h) > \eta_\alpha) = \mathbf{1}(\langle S(X') - S(X), h - S(X)\rangle_{\mathcal{H}} > c_\alpha)$$

Denote $f(h) = \langle S(X') - S(X), h - S(X)\rangle_{\mathcal{H}}$ as the test statistic.
Recall the definition/property of the Gaussian process, notice that $f(h)$ is a continuous linear functional of h, when $h \sim P_X \overset{d}{=} \mathcal{N}_{\mathbb{B}}(S(X), \sigma^2 C)$, it follows that $f(h)$ is a normal distribution in $\mathbb{R}$:

$$f(h) \sim N(f(0), \sigma^2 C(f, f)) \Longrightarrow f(x) \sim N(0, \sigma^2\|S(X) - S(X')\|_{\mathcal{H}}^2)$$

The last step is derived based on the Riesz representation theorem.
Based on the previous result, the Type I error of $\phi(h)$ is given by

$$\begin{aligned}\alpha_\phi =& \mathbb{E}_{P_X}[\phi(h)]\\ =& P_X(f(h) > c_\alpha)\\ =& 1 - P_X(f(h) \le c_\alpha)\\ =& 1 - P_X\left(\frac{f(h)}{\sigma\|S(X) - S(X')\|_{\mathcal{H}}} \le \frac{c}{\sigma\|S(X) - S(X')\|_{\mathcal{H}}}\right)\\ =& 1 - \Phi\left(\frac{c}{\sigma\|S(X) - S(X')\|_{\mathcal{H}}}\right)\end{aligned}$$

when $\alpha_\phi \le \alpha$, we have $\dfrac{c}{\sigma\|\theta_D - \theta_{D'}\|_{\mathcal{H}}} \ge \Phi^{-1}(1-\alpha)$
Similarly, we can get the type II error of the ϕ

$$\begin{aligned}
\beta_\phi =& 1 - \mathbb{E}_{P_{X'}}[\phi(h)] \\
=& 1 - P_{X'}(f(h) > c_\alpha) \\
=& P_{X'}(f(h) \le c_\alpha) \\
=& P_{X'}\left(\frac{f(h) - \|S(X) - S(X')\|_{\mathcal{H}}^2}{\sigma\|S(X) - S(X')\|_{\mathcal{H}}} \le \frac{c - \|S(X) - S(X')\|_{\mathcal{H}}^2}{\sigma\|S(X) - S(X')\|_{\mathcal{H}}}\right) \\
=& \Phi\left(\frac{c - \|S(X) - S(X')\|_{\mathcal{H}}^2}{\sigma\|S(X) - S(X')\|_{\mathcal{H}}}\right)
\end{aligned} \tag{13}$$

(13) is obtained since $f(h) \sim N(\|S(X) - S(X')\|_{\mathcal{H}}^2, \sigma^2\|S(X) - S(X')\|_{\mathcal{H}}^2)$
Thus, we get that

$$\begin{aligned}
T(P_X, P_{X'}) =& \inf_\phi \{\beta_\phi | \alpha_\phi \le \alpha\} \\
=& \Phi\left(\Phi^{-1}(1-\alpha) - \frac{\|S(X) - S(X')\|_{\mathcal{H}}}{\sigma}\right) \\
\ge& \Phi(\Phi^{-1}(1-\alpha) - \mu) = G_\mu
\end{aligned}$$

Therefore, $\widetilde{S}(X) = S(X) + \sigma Z$ satisfied μ-GDP. ■

Proof [Proof for Theorem 17] Since Φ has full column rank, $M = \Phi^\top \Phi$ is positive definite. Notice that the true coefficient vectors satisfy

$$\|\theta_i\|_2 \le B, \qquad i = 1, \dots, n.$$

Define the constraint set

$$\mathcal{C}_{B'} := \left\{\theta \in \mathbb{R}^m : \|\theta\|_2 \le B'\right\}.$$

For each individual i, first consider the unconstrained least-squares estimator

$$\hat{\theta}_i^{(u)} := \arg\min_{\theta \in \mathbb{R}^m} \|x_i - \Phi\theta\|_2^2.$$

We denote the following notations for convenience:

$$\begin{aligned}
e_i =& \hat{\theta}_i^{(c)} - \theta_i \\
\xi_i =& \hat{\theta}_i^{(u)} - \theta_i \\
r_i =& \hat{\theta}_i^{(c)} - \hat{\theta}_i^{(u)} \\
b_i =& \mathbb{E}\, e_i
\end{aligned} \tag{14}$$

It follows that

$$e_i = \xi_i + r_i$$

Since $M = \Phi^\top \Phi$ is positive definite,

$$\hat{\theta}_i^{(u)} = M^{-1}\Phi^\top x_i.$$

Using

$$x_i = \Phi\theta_i + \epsilon_i,$$

we obtain

$$\xi_i = \hat{\theta}_i^{(u)} - \theta_i = M^{-1}\Phi^\top \epsilon_i. \tag{15}$$

Because

$$\epsilon_i \sim N(0, \sigma_0^2 I_T),$$

it follows that

$$\xi_i \sim N(0, \sigma_0^2 M^{-1}) \tag{16}$$

Consequently,

$$\mathbb{E}\,\|\xi_i\|_2^2 = \operatorname{tr}\left[\operatorname{Cov}\left(\xi_i\right)\right] = \sigma_0^2 \operatorname{tr}(M^{-1}). \tag{17}$$

We next express the constrained estimator as a metric projection.

$$\hat{\theta}_i^{(c)} = \arg\min_{\theta \in \mathcal{C}_{B'}} \|x_i - \Phi\theta\|_2^2.$$

Using

$$\Phi^\top x_i = M\hat{\theta}_i^{(u)},$$

we can complete the square:

$$\begin{aligned}
\|x_i - \Phi\theta\|_2^2 &= x_i^\top x_i - 2\theta^\top \Phi^\top x_i + \theta^\top M\theta \\
&= x_i^\top x_i - 2\theta^\top M\hat{\theta}_i^{(u)} + \theta^\top M\theta \\
&= C_i + (\theta - \hat{\theta}_i^{(u)})^\top M(\theta - \hat{\theta}_i^{(u)}),
\end{aligned}$$

where $C_i = x_i^\top x_i - \hat{\theta}_i^{(u)\top} M\hat{\theta}_i^{(u)}$ does not depend on θ. Hence

$$\hat{\theta}_i^{(c)} = \arg\min_{\theta \in \mathcal{C}_{B'}} (\theta - \hat{\theta}_i^{(u)})^\top M(\theta - \hat{\theta}_i^{(u)}). \tag{18}$$

Define an inner product on $\mathbb{R}^m$ by

$$\langle u, v\rangle_M := u^\top M v, \qquad u, v \in \mathbb{R}^m,$$

and let

$$\|u\|_M := \sqrt{\langle u, u\rangle_M} = \sqrt{u^\top M u}$$

denote the corresponding norm. Since M is positive definite, $(\mathbb{R}^m, \langle\cdot,\cdot\rangle_M)$ is a Hilbert space. Equation (18) can therefore be written as

$$\hat{\theta}_i^{(c)} = \operatorname*{argmin}_{\theta \in \mathcal{C}_{B'}} \left\|\theta - \hat{\theta}_i^{(u)}\right\|_M^2 = P_{\mathcal{C}_{B'}}^M\left(\hat{\theta}_i^{(u)}\right), \tag{19}$$

where $P^M_{\mathcal{C}_{B'}}$ denotes the metric projection onto $\mathcal{C}_{B'}$ with respect to the M-norm.

The set $\mathcal{C}_{B'}$ is nonempty, closed, and convex. By Proposition 4.16 of Bauschke and Combettes (2017), the metric projection onto a nonempty closed convex subset of a Hilbert space is firmly nonexpansive, and hence nonexpansive. Thus, in the Hilbert space $(\mathbb{R}^m, \langle\cdot,\cdot\rangle_M)$,

$$\left\| P^M_{\mathcal{C}_{B'}}(u) - P^M_{\mathcal{C}_{B'}}(v) \right\|_M \le \|u - v\|_M \tag{20}$$

for all $u, v \in \mathbb{R}^m$.

Since $\|\theta_i\|_2 \le B \le B'$, we have $\theta_i \in \mathcal{C}_{B'}$. Therefore,

$$P^M_{\mathcal{C}_{B'}}(\theta_i) = \theta_i.$$

Applying (20) with

$$u = \hat{\theta}_i^{(u)}, \qquad v = \theta_i,$$

and using (19), we obtain

$$\begin{aligned} \|e_i\|_M &= \left\| \hat{\theta}_i^{(c)} - \theta_i \right\|_M \\ &= \left\| P^M_{\mathcal{C}_{B'}}\left(\hat{\theta}_i^{(u)}\right) - P^M_{\mathcal{C}_{B'}}(\theta_i) \right\|_M \\ &\le \left\| \hat{\theta}_i^{(u)} - \theta_i \right\|_M \\ &= \|\xi_i\|_M \end{aligned} \tag{21}$$

Let

$$\lambda_{\min}(M) \quad \text{and} \quad \lambda_{\max}(M)$$

denote the smallest and largest eigenvalues of M, respectively. Since M is positive definite, for every $v \in \mathbb{R}^m$,

$$\lambda_{\min}(M)\|v\|_2^2 \le \|v\|_M^2 \le \lambda_{\max}(M)\|v\|_2^2. \tag{22}$$

Consequently, (21) implies

$$\lambda_{\min}(M)\,\|e_i\|_2^2 \le \|e_i\|_M^2 \le \|\xi_i\|_M^2 \le \lambda_{\max}(M)\,\|\xi_i\|_2^2\,.$$

Therefore,

$$\|e_i\|_2^2 \le \kappa(M)\,\|\xi_i\|_2^2\,, \tag{23}$$

where

$$\kappa(M) := \frac{\lambda_{\max}(M)}{\lambda_{\min}(M)}$$

is the spectral condition number of M.
Taking expectations in (23) and applying (17), we obtain

$$\mathbb{E}\,\|e_i\|_2^2 \le \kappa(M)\sigma_0^2\,\mathrm{tr}(M^{-1}) \tag{24}$$

Denote $C = \kappa(M)\sigma_0^2 \operatorname{tr}(M^{-1})$, notice that e_i's are independent

$$\begin{aligned}\mathbb{E}\|d\|_2^2 &= \frac{1}{n^2}\mathbb{E}\left\|\sum_{i=1}^{n} e_i\right\|_2^2 \\ &= \frac{1}{n^2}\mathbb{E}\left(\sum_{i=1}^{n}\|e_i\|_2^2 + \sum_{i\neq j} e_i^\top e_j\right) \\ &= \frac{1}{n^2}\left(\sum_{i=1}^{n}\mathbb{E}\|e_i\|_2^2 + \sum_{i\neq j}\mathbb{E}e_i^\top \mathbb{E}e_j\right)\end{aligned} \tag{25}$$

Recall Equation (24), it follows that

$$\frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}\|e_i\|_2^2 \leq \frac{\kappa(M)\sigma_0^2 \operatorname{tr}(M^{-1})}{n} \tag{26}$$

Then, we only need to bound the clipping bias term.
Recall the definition of $\hat{\theta}_i^{(c)}$ given by Equation (19), we have

$$\|r_i\|_M = \left\|\hat{\theta}_i^{(c)} - \hat{\theta}_i^{(u)}\right\|_M \leq \left\|\theta_i - \hat{\theta}_i^{(u)}\right\|_M = \|\xi_i\|_M \tag{27}$$

Combine with Equation (22), we have

$$\|\hat{\theta}_i^{(c)} - \hat{\theta}_i^{(u)}\|_2^2 \leq \kappa(M)\|\hat{\theta}_i^{(u)} - \theta_i\|_2^2 \tag{28}$$

Since $\hat{\theta}_i^{(u)}$ is the unconstrained minimizer, we know that $r_i = 0$ if $\|\hat{\theta}_i^{(u)}\|_2 \leq B'$.
Therefore, we can rewrite

$$r_i = \left(\hat{\theta}_i^{(c)} - \hat{\theta}_i^{(u)}\right)\cdot \mathbf{1}\left\{\|\hat{\theta}_i^{(u)}\|_2 > B'\right\}$$

It follows that

$$\begin{aligned}\|r_i\|_2^2 =& \|\hat{\theta}_i^{(c)} - \hat{\theta}_i^{(u)}\|_2^2 \cdot \mathbf{1}\left\{\|\hat{\theta}_i^{(u)}\|_2 > B'\right\} \\ \leq& \kappa(M)\|\hat{\theta}_i^{(u)} - \theta_i\|_2^2 \cdot \mathbf{1}\left\{\|\hat{\theta}_i^{(u)}\|_2 > B'\right\}\end{aligned} \tag{29}$$

Thus, we can obtain

$$\|b_i\|_2^2 = \|\mathbb{E}e_i\|_2^2 = \|\mathbb{E}r_i\|_2^2 \leq \mathbb{E}\|r_i\|_2^2 \leq \kappa(M)\mathbb{E}\left[\|\hat{\theta}_i^{(u)} - \theta_i\|_2^2 \cdot \mathbf{1}\left\{\|\hat{\theta}_i^{(u)}\|_2 > B'\right\}\right] \tag{30}$$

By the Cauchy–Schwarz inequality, we have

$$\begin{aligned}\mathbb{E}\left[\|\hat{\theta}_i^{(u)} - \theta_i\|_2^2 \cdot \mathbf{1}\left\{\|\hat{\theta}_i^{(u)}\|_2 > B'\right\}\right] &= \mathbb{E}\left[\|\xi_i\|_2^2 \cdot \mathbf{1}\left\{\|\hat{\theta}_i^{(u)}\|_2 > B'\right\}\right] \\ &\leq \sqrt{\mathbb{E}\|\xi_i\|_2^4}\sqrt{\Pr(\|\hat{\theta}_i^{(u)}\|_2 > B')}\end{aligned} \tag{31}$$

Since $\xi_i \overset{i.i.d.}{\sim} N(0,\Sigma)$ with $\Sigma = \sigma_0^2 M^{-1}$, we have

$$\sqrt{\mathbb{E}\|\xi_i\|_2^4} = \sqrt{(\operatorname{tr}\Sigma)^2 + 2\operatorname{tr}(\Sigma^2)} \le \sqrt{3(\operatorname{tr}\Sigma)^2} = \sqrt{3}\sigma_0^2 \operatorname{tr}(M^{-1}) \tag{32}$$

Recall $\|\theta_i\|_2 \le B$, denote $\delta = B' - B$, then $\left\{\|\hat{\theta}_i^{(u)}\|_2 > B'\right\}$ implies $\left\{\|\hat{\theta}_i^{(u)} - \theta_i\|_2 > \delta\right\}$, which leads to

$$\begin{aligned}\Pr(\|\hat{\theta}_i^{(u)}\|_2 > B') &\le \Pr(\|\hat{\theta}_i^{(u)} - \theta_i\|_2 > \delta) \\ &= \Pr(\|\xi_i\|_2 > \delta)\end{aligned} \tag{33}$$

Let $\xi_i = \sigma_0 M^{-1/2} Z_i$, $Z_i \sim N(0, I_m)$, $\|Z_i\|_2^2 \sim \chi_m^2$, we have

$$\|\xi_i\|_2^2 = \sigma_0^2 Z_i^\top M^{-1} Z_i \le \sigma_0^2 \lambda_{\max}(M^{-1}) \|Z_i\|_2^2 = \frac{\sigma_0^2}{\lambda_{\min}(M)} \|Z_i\|_2^2$$

Then, we can obtain

$$\Pr(\|\xi_i\|_2 > \delta) = \Pr(\|\xi_i\|_2^2 > \delta^2) \le \Pr\left(\frac{\sigma_0^2}{\lambda_{\min}(M)} \|Z_i\|_2^2 > \delta^2\right) = \Pr\left(\chi_m^2 > \frac{\delta^2 \lambda_{\min}(M)}{\sigma_0^2}\right)$$

By Lemma 1 from Laurent and Massart (2000),

$$\Pr(\chi_m^2 > (\sqrt{m} + \sqrt{2x})^2) \le \Pr(\chi_m^2 > m + 2\sqrt{mx} + 2x) \le \exp(-x) \tag{34}$$

By choosing $\sqrt{m} + \sqrt{2x} = \dfrac{\delta\sqrt{\lambda_{\min}(M)}}{\sigma_0}$ (when $\delta > \dfrac{\sqrt{m}\sigma_0}{\sqrt{\lambda_{\min}(M)}}$), we have

$$\begin{aligned}\Pr(\|\hat{\theta}_i^{(u)}\|_2 > B') &\le \Pr(\|\xi_i\|_2 > \delta) \\ &\le \Pr\left(\chi_m^2 > \frac{\delta^2 \lambda_{\min}(M)}{\sigma_0^2}\right) \\ &\le \exp\left[-\frac{1}{2}\left(\frac{\delta\sqrt{\lambda_{\min}(M)}}{\sigma_0} - \sqrt{m}\right)^2\right]\end{aligned} \tag{35}$$

Combine Equation (30), (31), (32) and (35), we obtain

$$\begin{aligned}\|b_i\|_2^2 &\le \kappa(M)\mathbb{E}\left[\|\hat{\theta}_i^{(u)} - \theta_i\|_2^2 \cdot \mathbf{1}\left\{\|\hat{\theta}_i^{(u)}\|_2 > B'\right\}\right] \\ &\le \kappa(M)\sqrt{\mathbb{E}\|\xi_i\|_2^4}\sqrt{\Pr(\|\hat{\theta}_i^{(u)}\|_2 > B')} \\ &\le \kappa(M)\sqrt{3}\sigma_0^2 \operatorname{tr}(M^{-1}) \exp\left[-\frac{1}{4}\left(\frac{\delta\sqrt{\lambda_{\min}(M)}}{\sigma_0} - \sqrt{m}\right)^2\right]\end{aligned} \tag{36}$$

By the Cauchy–Schwarz inequality, we have

$$\frac{1}{n^2}\sum_{i\neq j} \mathbb{E}e_i^\top \mathbb{E}e_j = \frac{1}{n^2}\sum_{i\neq j} b_i^\top b_j \le \left(1 - \frac{1}{n}\right)\max_i \|b_i\|_2^2 \tag{37}$$

Combine Equation (25), (26), (36) and (37), we get

$$\boxed{\mathbb{E}\|d\|_2^2 \leq \frac{\kappa(M)\sigma_0^2\,\mathrm{tr}(M^{-1})}{n} + \left(1-\frac{1}{n}\right)\kappa(M)\sqrt{3}\sigma_0^2\,\mathrm{tr}(M^{-1})\exp\left[-\frac{1}{4}\left(\frac{\delta\sqrt{\lambda_{\min}(M)}}{\sigma_0}-\sqrt{m}\right)^2\right]} \tag{38}$$

Finally, since

$$\widetilde{\theta}_{FTP} = \overline{\hat{\theta}^{(c)}} + Z = \bar{\theta} + d + Z,$$

and

$$\int_0^1 \phi(t)\phi(t)^\top\,dt = I_m,$$

we have

$$\begin{aligned}\mathrm{IMSE}_{FTP} &= \mathbb{E}\int_0^1 \left[\widetilde{\theta}_{FTP}^\top\phi(t) - \bar{\theta}^\top\phi(t)\right]^2 dt \\ &= \mathbb{E}\|d+Z\|_2^2.\end{aligned}$$

Because $Z \sim N(0,\sigma^2 I_m)$ is independent of d and $\mathbb{E}[Z]=0$,

$$\mathbb{E}\|d+Z\|_2^2 = \mathbb{E}\|d\|_2^2 + \mathbb{E}\|Z\|_2^2 = \mathbb{E}\|d\|_2^2 + m\sigma^2.$$

With

$$\sigma = \frac{2B'}{n\mu},$$

we have

$$m\sigma^2 = \frac{4mB'^2}{n^2\mu^2}.$$

Combining this identity with (38) yields

$$\boxed{\begin{aligned}\mathrm{IMSE}_{FTP} \leq &\frac{4mB'^2}{n^2\mu^2} + \frac{\kappa(M)\sigma_0^2\,\mathrm{tr}(M^{-1})}{n} \\ &+ \left(1-\frac{1}{n}\right)\kappa(M)\sqrt{3}\sigma_0^2\,\mathrm{tr}(M^{-1})\exp\left[-\frac{1}{4}\left(\frac{(B'-B)\sqrt{\lambda_{\min}(M)}}{\sigma_0}-\sqrt{m}\right)^2\right]\end{aligned}} \tag{39}$$

■

Proof [Proof for Theorem 18] Denote $c_j = \|\phi(t_j)\|_2$, when $\sigma_0 = 0$, $x_i = \Phi\theta_i$. Since

$$\|\theta_i\|_2 \leq B < B', i = 1,\ldots,n$$

we have

$$|x_{ij}| = |\phi(t_j)^\top\theta_i| \leq Bc_j < B'c_j = C_j.$$

Therefore the clipping $\mathrm{clip}(x_{ij}, C_j)$ is inactive and

$$\bar{x}^{(\mathrm{clip})} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \Phi\bar{\theta}.$$

It follows that the released private vector can be written as

$$\widetilde{x} = \Phi\bar{\theta} + Z,$$

where

$$Z \sim N(0, \tau^2 W^{-1}), \quad \tau^2 = \frac{4B'^2T}{n^2\mu^2}, \quad W = \mathrm{diag}\{c_1^{-2}, \ldots, c_T^{-2}\}$$

We first analyze the corresponding unconstrained WLS estimator. By definition,

$$\widetilde{\theta}^{(u)}_{\mathrm{PTF}} = A^{-1}\Phi^\top W\widetilde{x}.$$

Substituting $\widetilde{x} = \Phi\bar{\theta} + Z$ gives

$$\widetilde{\theta}^{(u)}_{\mathrm{PTF}} = \bar{\theta} + A^{-1}\Phi^\top WZ.$$

Since

$$\mathrm{Cov}(Z) = \tau^2 W^{-1},$$

we obtain

$$\begin{aligned}\mathrm{Cov}\left(\widetilde{\theta}^{(u)}_{\mathrm{PTF}}\right) &= A^{-1}\Phi^\top W(\tau^2 W^{-1})W\Phi A^{-1} \\ &= \tau^2 A^{-1}\Phi^\top W\Phi A^{-1} \\ &= \tau^2 A^{-1}.\end{aligned}$$

Hence

$$\widetilde{\theta}^{(u)}_{\mathrm{PTF}} - \bar{\theta} \sim N(0, \tau^2 A^{-1})$$

and therefore

$$\mathbb{E}\left\|\widetilde{\theta}^{(u)}_{\mathrm{PTF}} - \bar{\theta}\right\|_2^2 = \tau^2\,\mathrm{tr}(A^{-1}). \tag{40}$$

We next account for the constraint $\|\theta\|_2 \le B'$. Let

$$\mathcal{A} = \{\|\widetilde{\theta}^{(u)}_{\mathrm{PTF}}\|_2 > B'\}$$

denote the event that the unconstrained estimator violates the constraint. On the complementary event $\mathcal{A}^c$, the unconstrained estimator is feasible and hence

$$\widetilde{\theta}^{(c)}_{\mathrm{PTF}} = \widetilde{\theta}^{(u)}_{\mathrm{PTF}}$$

Therefore

$$\begin{aligned}\mathbb{E}\|\widetilde{\theta}^{(c)}_{\mathrm{PTF}} - \bar{\theta}\|_2^2 &= \mathbb{E}[\|\widetilde{\theta}^{(u)}_{\mathrm{PTF}} - \bar{\theta}\|_2^2\mathbf{1}_{\mathcal{A}^c}] + \mathbb{E}[\|\widetilde{\theta}^{(c)}_{\mathrm{PTF}} - \bar{\theta}\|_2^2\mathbf{1}_{\mathcal{A}}] \\ &\ge \mathbb{E}[\|\widetilde{\theta}^{(u)}_{\mathrm{PTF}} - \bar{\theta}\|_2^2\mathbf{1}_{\mathcal{A}^c}] \\ &= \mathbb{E}\|\widetilde{\theta}^{(u)}_{\mathrm{PTF}} - \bar{\theta}\|_2^2 - \mathbb{E}[\|\widetilde{\theta}^{(u)}_{\mathrm{PTF}} - \bar{\theta}\|_2^2\mathbf{1}_{\mathcal{A}}].\end{aligned}$$

Thus, by Equation (40),

$$\mathrm{IMSE}_{\mathrm{PTF}} \geq \tau^2 \operatorname{tr}(A^{-1}) - R_{\mathrm{PTF}}, \tag{41}$$

where

$$R_{\mathrm{PTF}} = \mathbb{E}[\|\widetilde{\theta}^{(u)}_{\mathrm{PTF}} - \bar{\theta}\|_2^2 \mathbf{1}_{\mathcal{A}}].$$

It remains to bound the remainder term. Let

$$\xi = \widetilde{\theta}^{(u)}_{\mathrm{PTF}} - \bar{\theta}.$$

Then

$$\xi \sim N(0, \tau^2 A^{-1}).$$

Since

$$\|\bar{\theta}\|_2 \leq B,$$

the event $\mathcal{A}$ implies

$$\|\xi\|_2 = \|\widetilde{\theta}^{(u)}_{\mathrm{PTF}} - \bar{\theta}\|_2 \geq \|\widetilde{\theta}^{(u)}_{\mathrm{PTF}}\|_2 - \|\bar{\theta}\|_2 > B' - B.$$

Consequently,

$$\mathcal{A} \subseteq \left\{ \|\xi\|_2 > B' - B \right\}.$$

Write

$$\xi = \tau A^{-1/2} z, \qquad z \sim N(0, I_m).$$

Since

$$\|A^{-1/2} z\|_2^2 \leq \lambda_{\max}(A^{-1}) \|z\|_2^2 = \frac{\|z\|_2^2}{\lambda_{\min}(A)},$$

we obtain

$$\Pr(\mathcal{A}) \leq \Pr\left(\|z\|_2^2 > \frac{(B' - B)^2 \lambda_{\min}(A)}{\tau^2} \right).$$

Because $\|z\|_2^2 \sim \chi^2_m$, apply the chi-square tail bound in Equation (34), we get

$$\boxed{\Pr(\mathcal{A}) \leq \exp\left[-\frac{1}{2} \left\{ \frac{(B' - B)\sqrt{\lambda_{\min}(A)}}{\tau} - \sqrt{m} \right\}^2 \right].} \tag{42}$$

Finally, by the Cauchy–Schwarz inequality,

$$R_{\mathrm{PTF}} \leq \left(\mathbb{E}\|\xi\|_2^4 \right)^{1/2} \Pr(\mathcal{A})^{1/2}.$$

For a centered Gaussian vector with covariance $\tau^2 A^{-1}$,

$$\mathbb{E}\|\xi\|_2^4 = \tau^4 \left[\{\operatorname{tr}(A^{-1})\}^2 + 2 \operatorname{tr}(A^{-2}) \right] \leq \tau^4 [3\{\operatorname{tr}(A^{-1})\}^2]$$

Therefore

$$\left(\mathbb{E}\|\xi\|_2^4 \right)^{1/2} \leq \tau^2 \sqrt{3} \operatorname{tr}(A^{-1})$$

Combining this inequality with Equation (42) and Equation (41) gives Equation (9). ■

Proof [Proof for Corollary 19] Define $u_j = \dfrac{\phi(t_j)}{\|\phi(t_j)\|_2} \in \mathbb{R}^{m \times 1}$, $A_j = u_j u_j^T$

A_j's are rank-1 orthogonal projectors such that

- $A_j \succeq 0$, $A_j^2 = A_j$, $A_j^T = A_j$
- $\text{trace}(A_j) = \text{trace}(u_j^T u_j) = 1$

Then, we can rewrite $S_T = \frac{1}{T}\sum_{j=1}^{T} A_j$, and it follows that $\text{trace}(S_T) = 1$
Assume that the eigenvalues of S_T are $\lambda_1, \cdots, \lambda_m > 0$ ($S_T \succeq 0$ & invertible)
Then, the eigenvalues of S_T^{-1} will be $\frac{1}{\lambda_1}, \cdots, \frac{1}{\lambda_m}$, hence,

$$\begin{aligned} R(m,T) &= \frac{m}{\text{trace}(S_T)^{-1}} \\ &= \frac{m}{\sum_{k=1}^{m} \frac{1}{\lambda_k}} \\ &\leq \frac{\sum_{k=1}^{m} \lambda_k}{m} \\ &= \frac{\text{trace}(S_T)}{m} = \frac{1}{m} \end{aligned} \tag{43}$$

The last inequality is based on the harmonic mean - arithmetic mean (HM-AM) inequality, and the equality hold only when λ_k's are equal. ■

Proof [Proof of Theorem 20] Throughout the proof, let

$$\mathbf{f}_i = (f_i(t_1), \ldots, f_i(t_T))^\top, \quad \bar{\mathbf{f}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{f}_i$$

Since K_T is positive definite, K_T^{-1} and $K_T^{1/2}$ are well-defined.
a. We first show that for any $f_i \in \mathcal{H}_K$ with $\|f_i\|_{\mathcal{H}_K} \leq B$, we can always find another function g_i in the span of $\{K(t_1,t), \ldots, K(t_T,t)\}$ such that

$$f_i(t_j) = g_i(t_j), \ \|g_i\|_{\mathcal{H}_K} \leq B$$

For each i, consider the minimum-norm interpolation problem

$$g_i = \underset{g \in \mathcal{H}_K}{\text{argmin}} \{\|g\|_{\mathcal{H}_K} : g(t_j) = f_i(t_j), \ j = 1, \ldots, T\}$$

By the representer theorem,

$$g_i(t) = \sum_{j=1}^{T} (\gamma_i)_j K(t_j, t), \quad \gamma_i = K_T^{-1}\mathbf{f}_i$$

Hence

$$\|g_i\|_{\mathcal{H}_K}^2 = \gamma_i^\top K_T \gamma_i = \mathbf{f}_i^\top K_T^{-1}\mathbf{f}_i$$

Since f_i itself is feasible for the interpolation problem,

$$\|g_i\|_{\mathcal{H}_K} \leq \|f_i\|_{\mathcal{H}_K} \leq B$$

Therefore,

$$\mathbf{f}_i^\top K_T^{-1} \mathbf{f}_i \leq B^2 \tag{44}$$

Define

$$\beta_i := K_T^{-1/2} \mathbf{f}_i$$

Then Equation (44) gives

$$\|g_i\|_{\mathcal{H}_K} = \|\beta_i\|_2 \leq B \tag{45}$$

b. We then derive the form of the constrained and unconstrained solution to Equation

$$\hat{f}_i^{(c)} = \underset{\|f\|_{\mathcal{H}_K} \leq B'}{\operatorname{argmin}} \frac{1}{T} \sum_{j=1}^{T} [x_{ij} - f(t_j)]^2. \tag{46}$$

and perform coefficient transformation for the convenience of following proof.
By the representer theorem, the constrained estimator has the form

$$\hat{f}_i^{(c)}(t) = \sum_{j=1}^{T} (\hat{\alpha}_i^{(c)})_j K(t_j, t),$$

and its RKHS norm satisfies

$$\|\hat{f}_i^{(c)}\|_{\mathcal{H}_K}^2 = \hat{\alpha}_i^{(c)\top} K_T \hat{\alpha}_i^{(c)}$$

Consequently, Equation (46) is equivalent to

$$\hat{\alpha}_i^{(c)} = \underset{\alpha^\top K_T \alpha \leq B'^2}{\operatorname{argmin}} \|x_i - K_T \alpha\|_2^2 \tag{47}$$

Consider the corresponding unconstrained estimator

$$\hat{f}_i^{(u)} = \underset{f \in \mathcal{H}_K}{\operatorname{argmin}} \frac{1}{T} \sum_{j=1}^{T} [x_{ij} - f(t_j)]^2$$

The minimum-norm unconstrained least-squares solution has the form

$$\begin{aligned} \hat{f}_i^{(u)}(t) &= \sum_{j=1}^{T} [\hat{\alpha}_i^{(u)}]_j K(t_j, t), \\ \hat{\alpha}_i^{(u)} &= K_T^{-1} x_i \end{aligned} \tag{48}$$

Indeed, the fitted-value vector is

$$\hat{\mathbf{f}}_i^{(u)} = K_T \hat{\alpha}_i^{(u)} = x_i$$

Define

$$\beta_i^{(u)} := K_T^{1/2} \hat{\alpha}_i^{(u)} = K_T^{-1/2} x_i \quad \text{and} \quad \beta := K_T^{1/2} \alpha$$

We have

$$\alpha^\top K_T \alpha \leq B'^2 \iff \|\beta\|_2 = \|K_T^{1/2} \alpha\|_2 \leq B'$$

Hence, we can rewrite the optimization problem in Equation (47) as

$$\hat{\beta}_i^{(c)} = \underset{\|\beta\|_2 \le B'}{\operatorname{argmin}} \|x_i - K_T^{1/2}\beta\|_2^2, \quad \hat{\beta}_i^{(c)} = K_T^{1/2}\hat{\alpha}_i^{(c)} \tag{49}$$

It follows

$$\hat{f}_i^{(c)}(t) = \sum_{j=1}^{T} (\hat{\alpha}_i^{(c)})_j K(t_j, t) = \sum_{j=1}^{T} (K_T^{-1/2}\hat{\beta}_i^{(c)})_j K(t_j, t)$$

c. Next, we introduce the metric projection and follow similar steps in the proof of Theorem 17.
Denote

$$\mathcal{C}_{B'} := \left\{\beta \in \mathbb{R}^T : \|\beta\|_2 \le B'\right\}$$

By substituting $\Phi = K_T^{1/2}$, $M = \Phi^\top\Phi = \Phi^2 = K_T$, we have

$$\hat{\beta}_i^{(c)} = \underset{\beta\in\mathcal{C}_{B'}}{\operatorname{argmin}} (\beta - \hat{\beta}_i^{(u)})^\top M (\beta - \hat{\beta}_i^{(u)}) := P_{\mathcal{C}_{B'}}^M \left(\hat{\beta}_i^{(u)}\right) \tag{50}$$

where $P_{\mathcal{C}_{B'}}^M$ denotes the metric projection onto $\mathcal{C}_{B'}$ with respect to the M-norm.
Since the projection is nonexpansive, and $\|\beta_i\|_2 \le B \le B'$, we have

$$\left\|\hat{\beta}_i^{(c)} - \beta_i\right\|_M = \left\|P_{\mathcal{C}_{B'}}^M \left(\hat{\beta}_i^{(u)}\right) - P_{\mathcal{C}_{B'}}^M(\beta_i)\right\|_M \le \left\|\hat{\beta}_i^{(u)} - \beta_i\right\|_M \tag{51}$$

d. We denote convenient notations as follows

$$\begin{aligned} e_i &= \hat{\beta}_i^{(c)} - \beta_i \\ \xi_i &= \hat{\beta}_i^{(u)} - \beta_i \\ r_i &= \hat{\beta}_i^{(c)} - \hat{\beta}_i^{(u)} \\ b_i &= \mathbb{E}\, e_i \end{aligned} \tag{52}$$

Then Equation (51) becomes

$$\|e_i\|_M \le \|\xi_i\|_M \tag{53}$$

Since

$$x_i = \mathbf{f}_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma_0^2 I_T),$$

we have

$$\beta_i^{(u)} = K_T^{-1/2} x_i = \Phi^{-1} x_i = \beta_i + \xi_i,$$

where

$$\xi_i = \Phi^{-1}\epsilon_i, \quad \xi_i \sim N(0, \sigma_0^2 M^{-1})$$

Consequently,

$$\mathbb{E}\,\|\xi_i\|_2^2 = \operatorname{tr}\left[\operatorname{Cov}\left(\xi_i\right)\right] = \sigma_0^2 \operatorname{tr}(M^{-1}), \quad \mathbb{E}\,\|\epsilon_i\|_2^2 = \sigma_0^2 T \tag{54}$$

Then we can expand the MSE as

$$\begin{aligned}\mathrm{MSE}_{\mathrm{FTP}} &= \frac{1}{T}\sum_{j=1}^{T}\mathbb{E}[\widetilde{f}_{\mathrm{FTP}}(t_j) - \bar{f}(t_j)]^2 \\ &= \frac{1}{T}\mathbb{E}\|\widetilde{\mathbf{f}}_{\mathrm{FTP}} - \bar{\mathbf{f}}\|_2^2 \\ &= \frac{1}{T}\mathbb{E}\|\overline{\hat{\mathbf{f}}^{(c)}} - \bar{\mathbf{f}} + \sigma\mathbf{Z}\|_2^2 \end{aligned} \tag{55}$$

Notice that Z is a mean zero Gaussian process with kernel K, we have

$$\mathbf{Z} = (Z(t_1), \ldots, Z(t_T))^\top \sim \mathcal{N}(0, M)$$

$$\mathbb{E}\|\mathbf{Z}\|_2^2 = \mathrm{tr}(M)$$

Thus, we have

$$\begin{aligned}\mathrm{MSE}_{\mathrm{FTP}} &= \frac{1}{T}\mathbb{E}\|\overline{\hat{\mathbf{f}}^{(c)}} - \bar{\mathbf{f}} + \sigma\mathbf{Z}\|_2^2 \\ &= \frac{1}{T}\left(\mathbb{E}\|\overline{\hat{\mathbf{f}}^{(c)}} - \bar{\mathbf{f}}\|_2^2 + \mathbb{E}\|\sigma\mathbf{Z}\|_2^2 + 0\right) \\ &= \frac{1}{T}\mathbb{E}\|\overline{\hat{\mathbf{f}}^{(c)}} - \bar{\mathbf{f}}\|_2^2 + \frac{4B'^2}{n^2\mu^2 T}\,\mathrm{tr}(M) \end{aligned} \tag{56}$$

Note that e_i's are i.i.d, we can further expand the first term as

$$\begin{aligned}\mathbb{E}\|\overline{\hat{\mathbf{f}}^{(c)}} - \bar{\mathbf{f}}\|_2^2 &= \mathbb{E}\|\overline{\hat{\mathbf{f}}^{(c)}} - \bar{\mathbf{g}}\|_2^2 \\ &= \mathbb{E}\|\Phi\overline{\hat{\beta}^{(c)}} - \Phi\bar{\beta}\|_2^2 \\ &= \mathbb{E}\|\overline{\hat{\beta}^{(c)}} - \bar{\beta}\|_M^2 \\ &= \frac{1}{n^2}\mathbb{E}\left(\sum_{i=1}^{n}\|e_i\|_M^2 + \sum_{i\neq j} e_i^\top M e_j\right) \\ &= \frac{1}{n^2}\left(\sum_{i=1}^{n}\mathbb{E}\|e_i\|_M^2 + \sum_{i\neq j}\mathbb{E}e_i^\top M\mathbb{E}e_j\right) \end{aligned} \tag{57}$$

Recall Equation (53) and Equation (54), we have

$$\begin{aligned}\frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}\|e_i\|_M^2 &\leq \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}\|\xi_i\|_M^2 \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}\|\epsilon_i\|_2^2 \\ &= \frac{\sigma_0^2 T}{n} \end{aligned} \tag{58}$$

e. Finally, we analyze the cross-term.
Recall the definition of $\hat{\beta}_i^{(c)}$ given by Equation (50), we have

$$\|r_i\|_M = \left\|\hat{\beta}_i^{(c)} - \hat{\beta}_i^{(u)}\right\|_M \leq \left\|\beta_i - \hat{\beta}_i^{(u)}\right\|_M = \|\xi_i\|_M \tag{59}$$

Since $\hat{\beta}_i^{(u)}$ is the unconstrained minimizer, we know that $r_i = 0$ if $\|\hat{\beta}_i^{(u)}\|_2 \leq B'$.
Therefore, we can rewrite

$$r_i = \left(\hat{\beta}_i^{(c)} - \hat{\beta}_i^{(u)}\right) \cdot \mathbf{1}\left\{\|\hat{\beta}_i^{(u)}\|_2 > B'\right\}$$

It follows that

$$\begin{aligned}\|r_i\|_M^2 =& \|\hat{\beta}_i^{(c)} - \hat{\beta}_i^{(u)}\|_M^2 \cdot \mathbf{1}\left\{\|\hat{\beta}_i^{(u)}\|_2 > B'\right\} \\ \leq& \|\xi_i\|_M^2 \cdot \mathbf{1}\left\{\|\hat{\beta}_i^{(u)}\|_2 > B'\right\} \\ =& \|\epsilon_i\|_2^2 \cdot \mathbf{1}\left\{\|\hat{\beta}_i^{(u)}\|_2 > B'\right\}\end{aligned}$$

Thus, we can obtain that

$$\|b_i\|_M^2 = \|\mathbb{E}e_i\|_M^2 = \|\mathbb{E}r_i\|_M^2 \leq \mathbb{E}\|r_i\|_M^2 \leq \mathbb{E}\left[\|\epsilon_i\|_2^2 \cdot \mathbf{1}\left\{\|\hat{\beta}_i^{(u)}\|_2 > B'\right\}\right] \tag{60}$$

By the Cauchy–Schwarz inequality, we have

$$\mathbb{E}\left[\|\epsilon_i\|_2^2 \cdot \mathbf{1}\left\{\|\hat{\beta}_i^{(u)}\|_2 > B'\right\}\right] \leq \sqrt{\mathbb{E}\|\epsilon_i\|_2^4}\sqrt{\Pr(\|\hat{\beta}_i^{(u)}\|_2 > B')} \tag{61}$$

Since $\epsilon_i \overset{i.i.d.}{\sim} N(0, \Sigma)$ with $\Sigma = \sigma_0^2 I_T$, we have

$$\sqrt{\mathbb{E}\|\epsilon_i\|_2^4} = \sqrt{(\operatorname{tr}\Sigma)^2 + 2\operatorname{tr}(\Sigma^2)} = \sigma_0^2\sqrt{T^2 + 2T} \tag{62}$$

Recall $\|\beta_i\|_2 \leq B$, denote $\delta = B' - B$, then $\left\{\|\hat{\beta}_i^{(u)}\|_2 > B'\right\}$ implies $\left\{\|\hat{\beta}_i^{(u)} - \beta_i\|_2 > \delta\right\}$, thus

$$\begin{aligned}\Pr(\|\hat{\beta}_i^{(u)}\|_2 > B') &\leq \Pr(\|\hat{\beta}_i^{(u)} - \beta_i\|_2 > \delta) \\ &= \Pr(\|\xi_i\|_2 > \delta)\end{aligned}$$

Let $\xi_i = \sigma_0 M^{-1/2} Z_i$, $Z_i \sim N(0, I_T)$, $\|Z_i\|_2^2 \sim \chi_T^2$, we have

$$\|\xi_i\|_2^2 = \sigma_0^2 Z_i^\top M^{-1} Z_i \leq \sigma_0^2 \lambda_{\max}(M^{-1})\|Z_i\|_2^2 = \frac{\sigma_0^2}{\lambda_{\min}(M)}\|Z_i\|_2^2$$

Apply the χ^2 tail bound from Laurent and Massart (2000), we have

$$\begin{aligned}\Pr(\|\hat{\beta}_i^{(u)}\|_2 > B') &\leq \Pr(\|\xi_i\|_2 > \delta) \\ &\leq \Pr\left(\chi_T^2 > \frac{\delta^2\lambda_{\min}(M)}{\sigma_0^2}\right) \\ &\leq \exp\left[-\frac{1}{2}\left(\frac{\delta\sqrt{\lambda_{\min}(M)}}{\sigma_0} - \sqrt{T}\right)^2\right]\end{aligned} \tag{63}$$

Combine Equation (60), (61), (62) and (63), we obtain

$$\|b_i\|_M^2 \le \sigma_0^2\sqrt{T^2+2T}\cdot\exp\left[-\frac{1}{4}\left(\frac{\delta\sqrt{\lambda_{\min}(M)}}{\sigma_0}-\sqrt{T}\right)^2\right] \tag{64}$$

By the Cauchy–Schwarz inequality, we have

$$\frac{1}{n^2}\sum_{i\neq j}\mathbb{E}e_i^\top M\mathbb{E}e_j = \frac{1}{n^2}\sum_{i\neq j}b_i^\top Mb_j \le \left(1-\frac{1}{n}\right)\max_i\|b_i\|_M^2 \tag{65}$$

Finally combine Equation (56), (57), (58), (64), (65), and recall $K_T = M$, we get

$$\boxed{\begin{aligned}\mathrm{MSE}_{\mathrm{FTP}} \le & \frac{4B'^2\,\mathrm{tr}(K_T)}{n^2\mu^2T}+\frac{\sigma_0^2}{n}\\ &+\left(1-\frac{1}{n}\right)\sigma_0^2\sqrt{\frac{T+2}{T}}\exp\left[-\frac{1}{4}\left(\frac{(B'-B)\sqrt{\lambda_{\min}(K_T)}}{\sigma_0}-\sqrt{T}\right)^2\right]\end{aligned}} \tag{66}$$

Remark 27 (Minimum-norm interpolator problem) *Although both the FTP and PTF procedures involve the constrained optimization problems in Equations* (2) *and* (3)*, the role of the constraint is different in the noiseless setting. For the individual trajectories, when* $\sigma_0 = 0$ *we have* $x_{ij} = f_i(t_j)$*. Hence, the empirical least-squares loss can attain its global minimum value zero. In particular, the constrained problem*

$$\hat{f}_i = \operatorname*{argmin}_{\substack{f\in\mathcal{H}_K\\ \|f\|_{\mathcal{H}_K}\le B'}}\frac{1}{T}\sum_{j=1}^{T}[f(t_j)-x_{ij}]^2$$

admits f_i *as a feasible zero-loss solution. Consequently, the constraint does not affect the optimal value of the individual fitting problem.*

A convenient representative of the zero-loss solutions is the minimum-norm interpolant

$$\hat{f}_i^{\mathrm{MN}} = \operatorname*{argmin}_{f\in\mathcal{H}_K}\|f\|_{\mathcal{H}_K} \quad \textit{subject to} \quad f(t_j)=x_{ij}, \qquad j=1,\dots,T. \tag{67}$$

Let

$$K_T = [K_T(t_j,t_l)]_{j,l=1}^T$$

be the kernel Gram matrix, which is positive definite for distinct sampling locations under the Gaussian kernel. A direct Hilbert-space projection argument shows that the minimum-norm interpolation belongs to the finite-dimensional span of the kernel sections:

$$\hat{f}_i^{\mathrm{MN}}(t) = k_T(t)^\top K_T^{-1}x_i, \qquad k_T(t) = (K(t,t_1),\dots,K(t,t_T))^\top.$$

Moreover,

$$\|\hat{f}_i^{\mathrm{MN}}\|_{\mathcal{H}_K}^2 = x_i^\top K_T^{-1}x_i.$$

Since f_i itself is a feasible interpolation and $\|f_i\|_{\mathcal{H}_K} \le B$, the minimum-norm property gives

$$\|\hat{f}_i^{\mathrm{MN}}\|_{\mathcal{H}_K} \le \|f_i\|_{\mathcal{H}_K} \le B.$$

Thus, $\hat{f}_i^{\mathrm{MN}}$ is also feasible for the constrained least-squares problem and achieves its minimum loss, namely zero. Consequently, in the noiseless setting, we may take

$$\mathcal{F}_{\mathrm{FTP}}(x_i) = \hat{f}_i^{\mathrm{MN}}.$$

This observation allows the individual fitting step in the FTP procedure to be represented in closed form through the kernel Gram matrix.

■

Proof [Proof for Theorem 21] For each i, we can still define the minimum norm interpolant as

$$g_i = \operatorname*{argmin}_{g \in \mathcal{H}_K} \{\|g\|_{\mathcal{H}_K} : g(t_j) = f_i(t_j),\ j = 1, \dots, T\}$$

and its transformed coefficients as in the proof for Theorem 20:

$$\beta_i := K_T^{-1/2}\mathbf{f}_i, \quad \|g_i\|_{\mathcal{H}_K} = \|\beta_i\|_2 \le B$$

Denote $c_j = \sqrt{K(t_j, t_j)}$, by the RKHS reproducing property,

$$|f_i(t_j)| \le \|f_i\|_{\mathcal{H}_K}\sqrt{K(t_j, t_j)} \le Bc_j \le B'c_j$$

Since we are considering the noiseless case $\sigma_0 = 0$ and $x_i = \mathbf{f}_i$, with

$$C_j = B'c_j$$

the clipping operation $\mathrm{clip}(x_{ij}, C_j)$ is inactive and

$$\bar{x}^{(\mathrm{clip})} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{\mathbf{f}}$$

It follows that the released private vector can be written as

$$\widetilde{x} = \bar{\mathbf{f}} + Z = K_T^{1/2}\bar{\beta} + Z,$$

where

$$Z \sim N(0, \tau^2 W^{-1}), \quad \tau^2 = \frac{4B'^2 T}{n^2\mu^2}, \quad W = \mathrm{diag}\{c_1^{-2}, \dots, c_T^{-2}\}$$

By the representer theorem, the constrained estimator has the form

$$\hat{f}_{\mathrm{PTF}}^{(c)}(t) = \sum_{j=1}^{T} (\hat{\alpha}^{(c)})_j K(t_j, t),$$

$$\hat{\alpha}^{(c)}_{\mathrm{PTF}} = \underset{\alpha^\top K_T \alpha \le B'^2}{\operatorname{argmin}} \frac{1}{T}(\widetilde{x} - K_T\alpha)^\top W(\widetilde{x} - K_T\alpha)$$

Denote $\beta = K_T^{1/2}\alpha$, we can rewrite the constrained problem as

$$\hat{\beta}^{(c)}_{\mathrm{PTF}} = \underset{\|\beta\|_2 \le B'}{\operatorname{argmin}} (\widetilde{x} - K_T^{1/2}\beta)^\top W(\widetilde{x} - K_T^{1/2}\beta), \quad \hat{\beta}^{(c)}_{\mathrm{PTF}} = K_T^{1/2}\hat{\alpha}^{(c)}_{\mathrm{PTF}}$$

We then analyze the corresponding unconstrained WLS estimator.
Define

$$\Phi := K_T^{1/2}, \quad M := \Phi^\top\Phi = K_T, \quad A := \Phi^\top W\Phi = K_T^{1/2} W K_T^{1/2}$$

By definition, the unconstrained problem is:

$$\hat{\beta}^{(u)}_{\mathrm{PTF}} = \underset{\beta \in \mathbb{R}^T}{\operatorname{argmin}} (\widetilde{x} - K_T^{1/2}\beta)^\top W(\widetilde{x} - K_T^{1/2}\beta)$$

with the solution

$$\widetilde{\beta}^{(u)}_{\mathrm{PTF}} = A^{-1}\Phi^\top W\widetilde{x} = \Phi^{-1}\widetilde{x}$$

Substituting $\widetilde{x} = K_T^{1/2}\bar{\beta} + Z = \Phi\bar{\beta} + Z$ gives

$$\widetilde{\beta}^{(u)}_{\mathrm{PTF}} = \bar{\beta} + \xi$$

where

$$\xi := \Phi^{-1}Z$$

Since

$$Z \sim \mathcal{N}(0, \tau^2 W^{-1}),$$

we obtain

$$\xi \sim \mathcal{N}(0, \tau^2\Phi^{-1}W^{-1}\Phi^{-1}) \overset{d}{=} \mathcal{N}(0, \tau^2 A^{-1})$$

Hence

$$\mathbb{E}\,\|\xi\|_2^2 = \tau^2 \operatorname{tr}(A^{-1}), \quad \mathbb{E}\,\|Z\|_2^2 = \tau^2 \operatorname{tr}(W^{-1}) \tag{68}$$

We next account for the constraint $\|\beta\|_2 \le B'$. Let

$$\mathcal{A} = \{\|\widetilde{\beta}^{(u)}_{\mathrm{PTF}}\|_2 > B'\}$$

denote the event that the unconstrained estimator violates the constraint. On the complementary event $\mathcal{A}^c$, the unconstrained estimator is feasible and hence

$$\widetilde{\beta}^{(c)}_{\mathrm{PTF}} = \widetilde{\beta}^{(u)}_{\mathrm{PTF}}$$

Therefore

$$\begin{aligned}
\mathbb{E}\|\Phi(\widetilde{\beta}^{(c)}_{\mathrm{PTF}} - \bar{\beta})\|_2^2 &= \mathbb{E}[\|\Phi(\widetilde{\beta}^{(u)}_{\mathrm{PTF}} - \bar{\beta})\|_2^2 \mathbf{1}_{\mathcal{A}^c}] + \mathbb{E}[\|\Phi(\widetilde{\beta}^{(c)}_{\mathrm{PTF}} - \bar{\beta})\|_2^2 \mathbf{1}_{\mathcal{A}}] \\
&\ge \mathbb{E}[\|\Phi\xi\|_2^2 \mathbf{1}_{\mathcal{A}^c}] \\
&= \mathbb{E}\|Z\|_2^2 - \mathbb{E}[\|Z\|_2^2 \mathbf{1}_{\mathcal{A}}].
\end{aligned}$$

Thus, by Equation (68),

$$\begin{aligned} \mathrm{MSE}_{\mathrm{PTF}} &= \frac{1}{T}\mathbb{E}\|\widetilde{\mathbf{f}}_{\mathrm{PTF}} - \bar{\mathbf{f}}\|_2^2 \\ &= \frac{1}{T}\mathbb{E}\|\Phi(\widetilde{\beta}^{(c)}_{\mathrm{PTF}} - \bar{\beta})\|_2^2 \\ &\geq \frac{1}{T}\left[\tau^2 \operatorname{tr}(W^{-1}) - R_{\mathrm{PTF}}\right], \end{aligned} \tag{69}$$

where

$$R_{\mathrm{PTF}} = \mathbb{E}[\|Z\|_2^2 \mathbf{1}_{\mathcal{A}}].$$

It remains to bound the remainder term. Since

$$\|\bar{\beta}\|_2 \leq \frac{1}{n}\sum_{i=1}^{n}\|\beta_i\|_2 \leq B,$$

the event $\mathcal{A}$ implies

$$\|\xi\|_2 = \|\widetilde{\beta}^{(u)}_{\mathrm{PTF}} - \bar{\beta}\|_2 \geq \|\widetilde{\beta}^{(u)}_{\mathrm{PTF}}\|_2 - \|\bar{\beta}\|_2 > B' - B.$$

Consequently,

$$\mathcal{A} \subseteq \left\{\|\xi\|_2 > B' - B\right\}.$$

Write

$$\xi = \tau A^{-1/2} z, \qquad z \sim N(0, I_T).$$

Since

$$\|A^{-1/2}z\|_2^2 \leq \lambda_{\max}(A^{-1})\|z\|_2^2 = \frac{\|z\|_2^2}{\lambda_{\min}(A)},$$

we obtain

$$\Pr(\mathcal{A}) \leq \Pr\left(\|z\|_2^2 > \frac{(B'-B)^2\lambda_{\min}(A)}{\tau^2}\right).$$

Because $\|z\|_2^2 \sim \chi^2_T$, apply the chi-square tail bound in Equation (34), we get

$$\Pr(\mathcal{A}) \leq \exp\left[-\frac{1}{2}\left\{\frac{(B'-B)\sqrt{\lambda_{\min}(A)}}{\tau} - \sqrt{T}\right\}^2\right]. \tag{70}$$

Finally, by the Cauchy–Schwarz inequality,

$$R_{\mathrm{PTF}} \leq \left(\mathbb{E}\|Z\|_2^4\right)^{1/2} \Pr(\mathcal{A})^{1/2}.$$

For a centered Gaussian vector with covariance $\tau^2 W^{-1}$,

$$\mathbb{E}\|Z\|_2^4 = \tau^4\left[\{\operatorname{tr}(W^{-1})\}^2 + 2\operatorname{tr}(W^{-2})\right] \leq \tau^4[3\{\operatorname{tr}(W^{-1})\}^2]$$

Therefore

$$\left(\mathbb{E}\|Z\|_2^4\right)^{1/2} \leq \tau^2\sqrt{3}\operatorname{tr}(W^{-1})$$

Combining this inequality with Equation (70) and Equation (69), we get:

$$\boxed{\mathrm{MSE}_{\mathrm{PTF}} \geq \frac{\tau^2}{T}\,\mathrm{tr}(W^{-1})\left[1-\sqrt{3}\exp\left\{-\frac{1}{4}\left(\frac{(B'-B)\sqrt{\lambda_{\min}(A)}}{\tau}-\sqrt{T}\right)^2\right\}\right]} \tag{71}$$

■

Appendix C. Additional Simulation Results

Table 1: IMSE of FTP and PTF and their relative performance in Polynomial space with $\sigma_0 = 0$ across sample sizes n, basis dimension m, and GDP budget μ. The ratio is defined as $\mathrm{IMSE}_{\mathrm{FTP}}/\mathrm{IMSE}_{\mathrm{PTF}}$.

		$\mu = 0.5$			$\mu = 0.9$			$\mu = 1.3$		
n	B'	$\mathrm{IMSE}_{\mathrm{FTP}}$	$\mathrm{IMSE}_{\mathrm{PTF}}$	Ratio	$\mathrm{IMSE}_{\mathrm{FTP}}$	$\mathrm{IMSE}_{\mathrm{PTF}}$	Ratio	$\mathrm{IMSE}_{\mathrm{FTP}}$	$\mathrm{IMSE}_{\mathrm{PTF}}$	Ratio
100	B	0.066	0.139	0.476	0.062	0.132	0.473	0.063	0.132	0.475
	$\sqrt{\log}$	0.067	0.146	0.461	0.047	0.103	0.460	0.040	0.091	0.441
	log	0.142	0.261	0.546	0.068	0.115	0.592	0.052	0.075	0.685
800	B	0.083	0.196	0.423	0.084	0.196	0.428	0.083	0.195	0.425
	$\sqrt{\log}$	0.005	0.096	0.053	0.005	0.097	0.050	0.005	0.096	0.051
	log	0.008	0.031	0.266	0.006	0.027	0.208	0.005	0.027	0.198
6400	B	0.076	0.180	0.422	0.076	0.180	0.422	0.076	0.180	0.421
	$\sqrt{\log}$	0.001	0.078	0.008	0.001	0.078	0.008	0.001	0.078	0.008
	log	0.001	0.009	0.074	0.001	0.009	0.072	0.001	0.009	0.071

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